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### **KNOCKOUT OF A MUTANT ALLELE OF AN ELANE GENE**

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#### **Abstract**

Methods for inactivating in a cell a mutant allele of the elastase, neutrophil expressed gene (ELANE gene) gene having a mutation associated with severe congenital neutropenia (SCN) or cyclic neutropenia (CyN).

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## Background/Summary

[0001] This application claims the benefit of U.S. patent application Ser. No. 17/090,814 filed Nov. 5, 2020 and U.S. Prov. App. 62/931,655 filed Nov. 6, 2019. Both are hereby incorporated by reference.

[0002] Throughout this application, various publications are referenced, including referenced in parenthesis. The disclosures of all publications mentioned in this application in their entireties are hereby incorporated by reference into this application in order to provide additional description of the art to which this invention pertains and of the features in the art which can be employed with this invention.

### REFERENCE TO SEQUENCE LISTING

[0003] This application incorporates-by-reference nucleotide sequences which are present in the file named “5798\_0030US02\_KO\_ELANEv2.xml”, which is 39,683,353 bytes in size, and which was created on May 27, 2025 in the IBM-PC machine format, having an operating system compatibility with MS-Windows, which is contained in the xml file filed May 27, 2025 as part of this application.

### BACKGROUND OF INVENTION

[0004] A genetic disorder is caused by one or more abnormalities in the genome. Genetic disorders may be regarded as either “dominant” or “recessive.” Recessive genetic disorders are those which require two copies (i.e., two alleles) of the abnormal/defective gene to be present. In contrast, a dominant genetic disorder involves a gene or genes which exhibit(s) dominance over a normal (functional/healthy) gene or genes. As such, in dominant genetic disorders only a single copy (i.e., allele) of an abnormal gene is required to cause or contribute to the symptoms of a particular genetic disorder. Such mutations include, for example, gain-of-function mutations in which the altered gene product possesses a new molecular function or a new pattern of gene expression. Gain-of-function mutations are generally dominant negative mutations. An example of a dominant negative mutation is haploinsufficiency where one allele is mutated and loses its function and the single wild type allele left does not generate enough protein to be sufficient for a specific cellular function. Other examples include dominant negative mutations which have a gene product that acts antagonistically to the wild-type allele.

### Neutropenia

[0005] Neutropenia is defined as a reduction in the absolute number of neutrophils in the blood circulation and commonly diagnosed by measuring the absolute neutrophil count (ANC) in peripheral blood. The severity of neutropenia is characterized as mild with an ANC of 1000-1500/ $\mu$ L, moderate with an ANC of 500-1000/ $\mu$ L, or severe with an ANC of less than 500/ $\mu$ L (Boxer 2012).

[0006] Neutropenia can be classified as congenital (hereditary) or acquired. The two main types of the congenital condition, commonly of autosomal dominant inheritance, are cyclic neutropenia (CyN) and severe congenital neutropenia (SCN). Cyclic neutropenia is characterized by fluctuating neutrophil counts from normal levels to zero while severe congenital neutropenia (SCN) is characterized by very low ANC (500/ $\mu$ L) observed at birth, maturation arrest of the myelopoiesis in bone marrow at the promyelocyte/myelocyte stage, and early onset of bacterial infections (Carlsson et al. 2012; Horwitz et al. 2013).

[0007] SCN may be diagnosed by measuring a very low ANC in the blood and examining bone marrow aspirate to identify myeloid maturation arrest (Dale 2017). SCN is usually diagnosed

before age 6 months, while diagnosis for CyN is generally raised during the second year of life, or later, and the main clinical manifestation is recurrent acute stomatologic disorders. Bone marrow examination is often necessary to rule out malignant hemopathies, determine cellularity, assess myeloid maturation, and detect signs of a precise etiology, with cytogenetic bone marrow studies now crucial when SCN is suspected. Antineutrophil antibody assay, immunoglobulin assay (Ig GAM), lymphocyte immunophenotyping, pancreatic markers (serum trypsinogen and fecal elastase) and liposoluble vitamin levels (vitamins A, E and D) are also of interest in assessing SCN and CyN (See Donadieu 2011).

[0008] SCN can be autosomal-recessive (HAX1, G6PC3), autosomal-dominant (ELANE, GFI1), or X-linked (WAS) forms of inheritance or occur sporadically (Carlsson et al. 2012; Boxer 2012).

[0009] Cyclic and congenital neutropenia are most frequently caused by mutations in the “elastase, neutrophil expressed gene” (ELANE gene)—the gene for neutrophil elastase. “ELANE gene mutations are identified in 40-55% of SCN patients and males and females are equally affected (Donadieu et al. 2011; Dale 2017). Mutations in the ELANE gene are associated with autosomal-dominant and sporadic cases of SCN (Carlsson et al. 2012). To date, more than 200 different ELANE mutations have been identified, which are randomly distributed over all exons as well as in intron 3 and intron 4 (Skokowa et al. 2017). More than 120 distinct ELANE gene mutations related to CyN and SCN are now known, for example C151Y and G214R particularly associated with a poor prognosis. (See Makaryan et al. 2012; see also Germeshausen et al. 2013 for a comprehensive list of ELANE mutations related to CyN and SCN).

[0010] ELANE encodes neutrophil elastase (NE) which is involved in the function of neutrophil extracellular traps (networks of fibers that bind pathogens). Some studies suggest that the product of mutant ELANE acts to disrupt neutrophil production in the bone marrow and cause neutropenia. These studies indicate that mutations in NE initiate the unfolded protein response (UPR) leading to cell loss in the process of neutrophil formation in the marrow (Makaryan et al. 2017).

#### Current Treatments

[0011] Granulocyte colony-stimulating factor (G-CSF) is considered the first-line treatment for SCN (Connelly, Choi, and Levine 2012). G-CSF stimulates the production of more neutrophils and delays their apoptosis (Schäffer and Klein 2007). Overall survival is now estimated to exceed 80%, including patients developing malignancies, although 10% of SCN patients still die from severe bacterial infections or sepsis (Skokowa et al. 2017). Although G-CSF therapy is successful in preventing mortality from sepsis, long-term treatment was identified to be associated with an increased risk of developing myelodysplastic syndrome (MDS) or leukemia in SCN patients. The most common leukemia in SCN is AML, but acute lymphoid leukemia (ALL), juvenile myelomonocytic leukemia (JMML), chronic myelomonocytic leukemia (CMML), and bi-phenotypic leukemia are also reported in the literature (Connelly, Choi, and Levine 2012). It was previously demonstrated that patients who had a robust response to G-CSF (doses  $\leq 8 \mu\text{g/kg/day}$ ) had a cumulative incidence of 15% for developing MDS/leukemia after 15 years on G-CSF, while an incidence of 34% was reported in patients with poor response to G-CSF despite high doses (Rosenberg et al. 2010).

[0012] Hematopoietic stem cell transplant (HSCT) is an alternative, curative therapy for patients who do not respond to G-CSF therapy or who develop AML/MDS. However, patients with chronic neutropenia who undergo HCT are at increased risk of developing infectious complications such as fungal and graft-versus-host disease (Skokowa et al. 2017). Moreover, HCT requires a matched related donor for successful survival but most patients will not have an available matched donor (Connelly, Choi, and Levine 2012).

#### SUMMARY OF THE INVENTION

[0013] Disclosed is an approach for knocking out the expression of a dominant-mutant allele by disrupting the dominant-mutant allele or degrading the resulting mRNA.

[0014] Embodiments of the present invention provide methods for knocking out expression of a

mutant allele of the elastase, neutrophil expressed gene (ELANE gene) gene having a mutation associated with severe congenital neutropenia (SCN) or cyclic neutropenia (CyN) and allowing expression of the functional protein.

[0015] The present invention provides a method for inactivating in a cell a mutant allele of the elastase, neutrophil expressed gene (ELANE gene) gene having a mutation associated with severe congenital neutropenia (SCN) or cyclic neutropenia (CyN), wherein the mutant allele is selected from the group consisting of the ELANE mutants set forth in Table 1, the method comprising [0016] introducing to the cell a composition comprising: [0017] a CRISPR nuclease or a sequence encoding the CRISPR nuclease; and [0018] a first RNA molecule comprising a guide sequence portion having 17-30 nucleotides, [0019] wherein a complex of the CRISPR nuclease and the first RNA molecule affects a double strand break in the mutant allele of the ELANE gene.

[0020] The present invention provides for a modified cell obtained by the methods of the present invention.

[0021] The present invention provides for a modified cell lacking at least a portion of one allele of the ELANE gene.

[0022] The present invention provides for a composition comprising modified cells and a pharmaceutically acceptable carrier.

[0023] The present invention provides for an in vitro or ex vivo method of preparing a composition, comprising mixing the cells of the present invention with the pharmaceutically acceptable carrier.

[0024] The present invention provides for a method of preparing in vitro or ex vivo a composition comprising modified cells, the method comprising: [0025] a) isolating HSPCs from cells obtained from a subject with an ELANE gene mutation related to SCN or CyN and/or suffering from SCN or CyN and which subject has a mutant allele of the ELANE gene selected from the group consisting of the ELANE mutants set forth in Table 1, and obtaining the cell from the subject;

[0026] b) introducing to the cells of step (a) a composition comprising: [0027] a CRISPR nuclease or a sequence encoding the CRISPR nuclease; and [0028] a first RNA molecule comprising a guide sequence portion having 17-30 nucleotides, [0029] wherein a complex of the CRISPR nuclease and the first RNA molecule affects a double strand break in the mutant allele of the ELANE gene in one or more cells, [0030] optionally, introducing to the cells a second RNA molecule comprising a guide sequence portion capable of complexing with a CRISPR nuclease, wherein the complex of the second RNA molecule and CRISPR nuclease affects a second double strand break in the ELANE gene in the one or more cells [0031] so as to inactivate the mutant allele of the ELANE gene in one or more cells thereby obtaining modified cells; optionally [0032] c) culture expanding the modified cells of step (b), wherein the modified cells are capable of engraftment and giving rise to progeny cells after engraftment.

[0033] The present invention provides for use of a composition prepared in vitro by a method comprising: [0034] a) isolating HSPCs from cells obtained from a subject with an ELANE gene mutation related to SCN or CyN and/or suffering from SCN or CyN and subject has a mutant allele of the ELANE gene selected from the group consisting of the ELANE mutants set forth in Table 1; [0035] b) introducing to the cells of step (a) a composition comprising: [0036] a CRISPR nuclease or a sequence encoding the CRISPR nuclease; and [0037] a first RNA molecule comprising a guide sequence portion having 17-30 nucleotides, [0038] wherein a complex of the CRISPR nuclease and the first RNA molecule affects a double strand break in the mutant allele of the ELANE gene in one or more cells, [0039] optionally, introducing to the cells a second RNA molecule comprising a guide sequence portion capable of complexing with a CRISPR nuclease, wherein the complex of the second RNA molecule and CRISPR nuclease affects a second double strand break in the ELANE gene in the one or more cells [0040] so as to inactivate the mutant allele of the ELANE gene in one or more cells thereby obtaining modified cells; optionally; [0041] c) culture expanding the cells of step (b) wherein the modified cells are capable of engraftment and giving rise to progeny

cells after engraftment; and [0042] d) administering to the subject the cells of step (b) or step (c) [0043] for treating the SCN or CyN in the subject.

[0044] The present invention provides for a method of treating a subject afflicted with SCN or CyN, comprising administration of a therapeutically effective amount of the modified cells, compositions, or the compositions prepared by the methods of the instant invention

[0045] The present invention provides for a method for treating SCN or CyN in a subject with an ELANE gene mutation relating to SCN or CYN in need thereof and which subject has a mutant allele of the ELANE gene selected from the group consisting of the ELANE mutants set forth in Table 1, the method comprising: [0046] a) isolating HSPCs from cells obtained from the subject; [0047] b) introducing to the cells of step (a) a composition comprising: [0048] a CRISPR nuclease or a sequence encoding the CRISPR nuclease; and [0049] a first RNA molecule comprising a guide sequence portion having 17-30 nucleotides, wherein a complex of the CRISPR nuclease and the first RNA molecule affects a double strand break in the mutant allele of the ELANE gene in one or more cells, [0050] optionally, introducing to the cells a second RNA molecule comprising a guide sequence portion capable of complexing with a CRISPR nuclease, wherein the complex of the second RNA molecule and CRISPR nuclease affects a second double strand break in the ELANE gene in the one or more cells so as to inactivate the mutant allele of the ELANE gene in one or more cells thereby obtaining modified cells; optionally; [0051] c) culture expanding the cells of step (b) wherein the modified cells are capable of engraftment and giving rise to progeny cells after engraftment; and [0052] d) administering to the subject the cells of step (b) or step (c) [0053] thereby treating the SCN or CyN in the subject.

[0054] The present invention provides for a method for treating SCN or CyN in a subject with an ELANE gene mutation relating to SCN or CYN in need thereof and subject has a mutant allele of the ELANE gene selected from the group consisting of the ELANE mutants set forth in Table 1, the method comprising [0055] administering to the subject autologous modified cells or progeny of autologous modified cells, wherein the autologous modified cells are modified so as to have a double strand break in the mutant allele of the ELANE gene, [0056] wherein said double strand break results from introduction to the cells of a composition comprising a CRISPR nuclease or a sequence encoding the CRISPR nuclease and a first RNA molecule wherein a complex of the CRISPR nuclease and the first RNA molecule affects a double strand break in the mutant allele of the ELANE gene so as to inactivate the mutant allele of the ELANE gene in the cell, [0057] thereby treating the SCN or CyN in the subject.

[0058] The present invention provides for a method of selecting a subject for treatment from a pool of subjects diagnosed with SCN or CyN, comprising the steps of: [0059] a) obtaining cells from each subject in the pool of subjects; [0060] b) screening each subject's cells for an ELANE gene mutation related to SCN or CyN, and selecting only subjects with an ELANE gene mutation related to SCN or CyN; [0061] c) screening by sequencing the cells of the subjects selected in step (b) for a mutant allele of the ELANE gene selected from the group consisting of the ELANE mutations set forth in Table 1, [0062] d) selecting for treatment only subjects with cells heterozygous for an ELANE mutation set forth in Table 1 [0063] e) obtaining hematopoietic stem and progenitor cells (HSPC) cells from the bone marrow of the subject either by aspiration or by mobilization and apheresis of peripheral blood; and [0064] f) introducing to the HSPC cells of step (e): [0065] one or more CRISPR nucleases or sequences encoding the one or more CRISPR nuclease; [0066] a first RNA molecule comprising a guide sequence portion having 17-30 contiguous nucleotides containing nucleotides in the sequence set forth in any one of SEQ ID NOs: 1-31117 targeting the mutant allele of the ELANE gene, and [0067] a second RNA molecule comprising a guide sequence portion targeting a sequence in intron 3, intron 4 or 3' UTR of the ELANE gene, wherein a complex of the first RNA molecule and a CRISPR nuclease affects a first double strand break in the mutant allele of the ELANE gene in one or more of the HSPC cells and a complex of the second RNA molecule and a CRISPR nuclease affect a second double strand break in intron 3,

intron 4, or 3' UTR of both alleles of the ELANE gene in the one or more HSPC cells in which the complex of the first RNA molecule and the CRISPR nuclease affected a first double strand break, thereby obtaining modified cells; [0068] g) administering to the subject the modified cells of step (f), thereby treating SCN or CyN in the subject.

[0069] The present invention provides an RNA molecule comprising a guide sequence portion having 17-30 contiguous nucleotides containing nucleotides in the sequence set forth in any one of SEQ ID NOs: 1-31117.

[0070] The present invention provides a method for inactivating in a cell a mutant ELANE allele, the method comprising delivering to the cell the RNA molecules or compositions of the present invention.

[0071] The present invention provides use of the RNA molecules, the compositions, or the composition prepared by the method of the present invention for inactivating in a cell a mutant ELANE allele.

[0072] The present invention provides a medicament comprising the RNA molecules, compositions, or the compositions prepared by the methods of the instant invention for use in inactivating in a cell a mutant ELANE allele, wherein the medicament is administered by delivering to the cell the RNA molecules, compositions, or the compositions prepared by the methods of the instant invention.

[0073] The present invention provides for use of the methods, the modified cells, the compositions, or the compositions prepared by the methods, or the RNA molecules of the instant invention for treating ameliorating or preventing SCN or CyN in to a subject having or at risk of having SCN or CyN.

[0074] The present invention provides for a medicament comprising the RNA molecules, compositions, compositions prepared by the methods of the instant invention, or the modified cells of the instant invention, for use in treating ameliorating or preventing SCN or CyN, wherein the medicament is administered by delivering to a subject having or at risk of having SCN or CyN the RNA molecules, compositions, compositions prepared by the methods of the instant invention, or the modified cells of the instant invention.

[0075] The present invention provides for a kit for inactivating a mutant ELANE allele in a cell, comprising the RNA molecules of the instant invention, a CRISPR nuclease or a sequence encoding the CRISPR nuclease, and/or a tracrRNA molecule or a sequence encoding the tracrRNA; and instructions for delivering the RNA molecule; CRISPR nuclease or sequence encoding the CRISPR nuclease, and/or tracrRNA molecule or sequence encoding the tracrRNA to the cell to inactivate the mutant ELANE allele in the cell.

[0076] The present invention provides for a kit for treating SCN or CyN in a subject, comprising the RNA molecules of the instant invention, a CRISPR nuclease or sequence encoding the CRISPR nuclease, and/or a tracrRNA molecule or sequence encoding the tracrRNA molecule; and instructions for delivering the RNA molecule; CRISPR nuclease, and/or the tracrRNA to a subject having or at risk of having SCN or CyN so as to treat the SCN or CyN.

[0077] The present invention provides a kit for inactivating a mutant ELANE allele in a cell, comprising the compositions, the composition prepared by the methods of the instant invention, or the modified cells of the instant invention, and instructions for delivering the composition to the cell so as to inactivate the ELANE gene in the cell.

[0078] The present invention provides a kit for treating SCN or CyN in a subject, comprising the composition, the compositions prepared by the methods of the instant invention, or the modified cells of the instant invention, and instructions for delivering the compositions, the compositions prepared by the methods of the instant invention, or the modified cells of the instant invention, to a subject having or at risk of having SCN or CyN so as to treat SCN or CyN.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0079] FIG. 1: In-vitro PAM Depletion by TXTL results for OMNI-50. The OMNI-50 nuclease was challenged with two versions of sgRNA with targeting spacer (T2) and non-targeting spacer (T1). The PAM logo is a schematic representation of the ratio of the depleted site. A table with the tested sgRNA version is presented for the most depleted reads with their ratio comparing to the non-targeted library.

[0080] FIG. 2: Intrinsic fidelity in human cells. OMNI-50 or SpCas9 nuclease was expressed in a mammalian cell system by DNA transfection together with a sgRNA-expressing plasmid. Cell lysates were used for site specific genomic DNA amplification and NGS. The percentage of indels was measured and analyzed to compare on-target vs off-target editing in HeLa cell line using ELANEg35\_OMNI50 or ELANEg62\_OMNI50 (see Table 3). In both cases, the genomic On and Off target sequences are noted below the chart with the PAM sequence underlined. Each experiment represents three (3) independent repeats.

[0081] FIGS. 3A-3D: OMNI-50 activity Assay as RNP. OMNI-50 nuclease was over-expressed and purified. The purified protein was complexed with synthetic sgRNA to form RNPs. For the in-vitro assays (FIG. 3A and FIG. 3B) RNPs were incubated with a linear DNA template with the corresponding target and PAM sequences (listed in Table 5). Activity was verified by the ability to cleave the linear template. For the in-vivo assays (FIG. 3C and FIG. 3D) U2OS cells were electroporated with RNP, and activity was determined as indel frequency by NGS. FIG. 3A: Activity assay of OMNI-50 RNP with different spacer lengths (17-23 nucleotides) of guide 35 (Table 4).

[0082] FIG. 3B: Reducing amounts of RNPs (4, 2, 1.2, 0.6 and 0.2  $\mu$ mol) with spacer lengths 20-23 nucleotides were incubated with 100 ng of DNA target template. FIG. 3C: Activity assay for OMNI-50 as RNP in U2OS cells: RNPs with spacer lengths 17-23 nucleotides were electroporated into a U2OS cell line and editing levels (indels) were measured by NGS. FIG. 3D. Activity assay for OMNI-50 as RNP in U2OS cells: RNPs with ELANE g35 sgRNA V1-4 were electroporated into a U2OS cell line and editing levels (indels) were measured by NGS.

[0083] FIG. 4. Activity assay for OMNI-50 as RNP in iPSCs: RNPs with spacer lengths 17-23 nucleotides (Table 4) were electroporated into an iPSC cell line and editing levels (indels) were measured by NGS.

### DETAILED DESCRIPTION

[0084] Embodiments of the present invention provide a method for inactivating in a cell a mutant allele of the elastase, neutrophil expressed gene (ELANE gene) gene having a mutation associated with severe congenital neutropenia (SCN) or cyclic neutropenia (CyN) wherein the mutant allele is selected from the group consisting of the ELANE mutants set forth in Table 1, the method comprising [0085] introducing to the cell a composition comprising: [0086] a CRISPR nuclease or a sequence encoding the CRISPR nuclease; and [0087] a first RNA molecule comprising a guide sequence portion having 17-30 nucleotides, [0088] wherein a complex of the CRISPR nuclease and the first RNA molecule affects a double strand break in the mutant allele of the ELANE gene.

[0089] In embodiments of the present invention, the guide sequence portion of the first RNA molecule comprises 17-22 contiguous nucleotides as set forth in any one of SEQ ID NOs: 1-31117.

[0090] In embodiments of the present invention, the complex of the CRISPR nuclease and the first RNA molecule affects a double strand break in the mutant allele of the ELANE gene, wherein the mutation associated with severe congenital neutropenia (SCN) or cyclic neutropenia (CyN) in mutant allele is targeted for the double strand break by the complex of the CRISPR nuclease and the first RNA molecule.

[0091] In embodiments of the present invention, the CRISPR nuclease and the first RNA molecule

affects a double strand break in the mutant allele of the ELANE gene, which mutant allele is targeted for the double strand break based on a sequence of the mutant allele at the mutation associated with severe congenital neutropenia (SCN) or cyclic neutropenia (CyN).

[0092] In embodiments of the present invention, the CRISPR nuclease and the first RNA molecule affects a double strand break in the mutant allele of the ELANE based on the nucleotide base of the mutation associated with severe congenital neutropenia (SCN) or cyclic neutropenia (CyN) present on the mutant allele of the ELANE gene.

[0093] Embodiments of the present invention further comprise introduction of a second RNA molecule comprising a guide sequence portion capable of complexing with a CRISPR nuclease, wherein the complex of the second RNA molecule and CRISPR nuclease affects a second double strand break in the ELANE gene.

[0094] In embodiments of the present invention, a composition may comprise 1, 2, 3 or more CRISPR nucleases or sequencing encoding the CRISPR nucleases. In embodiments of the present invention, introducing a composition to the cell may comprise introducing 1, 2, 3, or more compositions to the cell. In embodiments of the present invention, each composition may comprise a different CRISPR nuclease or sequence encoding the CRISPR nucleases or the same CRISPR nuclease or sequence encoding the CRISPR nuclease. In embodiments of the present invention involving two RNA molecules, the second RNA molecule may form a complex with the same CRISPR nuclease as the first RNA molecule, or may form a complex with another CRISPR nuclease.

[0095] In embodiments of the present invention, the second double strand break is within a non-coding region of the ELANE gene. In embodiments of the present invention, the non-coding region of the ELANE gene is selected an intron or an untranslated region (UTR). In embodiments of the present invention, the non-coding region is in intron 3 or intron 4. In an embodiment of the present invention the UTR is the 3'UTR.

[0096] In embodiments of the present invention, the guide sequence portion of the first RNA molecule comprises 17-22 contiguous nucleotides as set forth in any one of SEQ ID NOs: 1-31117.

[0097] According to embodiments of the present invention, the guide sequence portion of the second RNA molecule comprises 17-30 contiguous nucleotides containing nucleotides in the sequence set forth in any one of SEQ ID NOs: 1-31117.

[0098] In embodiments of the present invention, the second double strand break is within a non-coding region of the ELANE gene.

[0099] Embodiments of the present invention comprise obtaining the cell with an ELANE gene mutation associated with severe congenital neutropenia (SCN) or CyN from a subject with an ELANE gene mutation related to SCN or CyN and/or suffering from SCN or CyN wherein the ELANE gene mutation is selected from the group consisting of the ELANE mutants set forth in Table 1.

[0100] Embodiments of the present invention comprise first selecting a subject with an ELANE gene mutation related to SCN or CyN and/or suffering from SCN or CyN wherein the ELANE gene mutation is selected from the group consisting of the ELANE mutants set forth in Table 1, and obtaining the cell from the subject.

[0101] Embodiments of the present invention comprise obtaining the cell from the subject by mobilization and/or by apheresis.

[0102] Embodiments of the present invention comprise obtaining the cell from the subject by bone marrow aspiration.

[0103] In embodiments of the present invention, the cell is prestimulated prior to introducing the composition to the cell.

[0104] Embodiments of the present invention comprise culture expanding the cell to obtain cells.

[0105] In embodiments of the present invention, the cells are cultured with one or more of: stem cell factor (SCF), IL-3, and GM-CSF.



[0106] In embodiments of the present invention, the cells are cultured with at least one cytokine.

[0107] In embodiments of the present invention, the at least one cytokine is a recombinant human cytokine.

[0108] In embodiments of the present invention, the cell is among a plurality of cells, wherein the composition comprising the first RNA molecule or both the first and the second RNA molecule is introduced into at least the cell as well as other cells among the plurality of cells, and the mutant allele of the ELANE gene is inactivated in at least the cell as well as in the other cells among the plurality of cells, thereby obtaining multiple modified cells.

[0109] In embodiments of the present invention, introducing the composition comprising the first RNA molecule or introduction of the second RNA molecule comprises electroporation of the cell or cells.

[0110] Embodiments of the present invention provide for a modified cell obtained by the methods of the present invention.

[0111] In embodiments of the present invention, the modified cells are further culture expanded.

[0112] In embodiments of the present invention, the modified cells are capable of engraftment.

[0113] In embodiments of the invention, modified cells are capable of long-term engraftment when infused into a patient, giving rise to differentiated hematopoietic cells for at least 12 months after infusion, preferably at least 24 months and even more preferably at least 30 months after infusion.

In a further embodiment, the modified cells are capable of long-term engraftment when infused into an autologous subject. In a further embodiment, the modified cells are capable of long-term engraftment when infused into a subject without myeloablation. In an embodiment of the present invention, the modified cells are delivered to a subject in sufficient numbers that, when engrafted into a human subject, provide long term engraftment.

[0114] In embodiments of the present invention, the modified cell or cells are capable of giving rise to progeny cells.

[0115] In embodiments of the present invention, the modified cell or cells are capable of giving rise to progeny cells after engraftment.

[0116] In embodiments of the present invention, the modified cell or cells are capable of giving rise to progeny cells after an autologous engraftment.

[0117] In embodiments of the present invention, the modified cell or cells are capable of giving rise to progeny cells for at least 12 months or at least 24 months after engraftment.

[0118] In one embodiment, the cell or cells are stem cells. In one embodiment, the cell is an embryonic stem cell. In some embodiment, the stem cell is a hematopoietic stem/progenitor cell (HSPC).

[0119] In embodiments of the present invention, the modified cell or cells are CD34+ hematopoietic stem cells.

[0120] In embodiments of the present invention, the modified cell or cells are bone marrow cells or peripheral mononucleated cells (PMCs).

[0121] Embodiments of the present invention provide for a modified cell lacking at least a portion of one allele of the ELANE gene.

[0122] In embodiments of the present invention, the modified cell was modified from a cell with an ELANE gene mutation selected from the group consisting of the ELANE mutants set forth in Table 1.

[0123] Embodiments of the present invention provide for a composition comprising modified cells and a pharmaceutically acceptable carrier.

[0124] Embodiments of the present invention provide for an in vitro or ex vivo method of preparing a composition, comprising mixing the cells of the present invention with the pharmaceutically acceptable carrier.

[0125] Embodiments of the present invention provide for a method of preparing in vitro or ex vivo a composition comprising modified cells, the method comprising: [0126] a) isolating HSPCs from

cells obtained from a subject with an ELANE gene mutation related to SCN or CyN and/or suffering from SCN or CyN, wherein the ELANE gene mutation is selected from the group consisting of the ELANE mutants set forth in Table 1, and obtaining the cell from the subject; [0127] b) introducing to the cells of step (a) a composition comprising: [0128] a CRISPR nuclease or a sequence encoding the CRISPR nuclease; and [0129] a first RNA molecule comprising a guide sequence portion having 17-30 nucleotides, wherein a complex of the CRISPR nuclease and the first RNA molecule affects a double strand break in the mutant allele of the ELANE gene in one or more cells, [0130] optionally, introducing to the cells a second RNA molecule comprising a guide sequence portion capable of complexing with a CRISPR nuclease, wherein the complex of the second RNA molecule and CRISPR nuclease affects a second double strand break in the ELANE gene in the one or more cells so as to inactivate the mutant allele of the ELANE gene in one or more cells thereby obtaining modified cells; optionally [0131] c) culture expanding the modified cells of step (b), [0132] wherein the modified cells are capable of engraftment and giving rise to progeny cells after engraftment.

[0133] Embodiments of the present invention provide for use of a composition prepared in vitro by a method comprising: [0134] a) isolating HSPCs from cells obtained from a subject with an ELANE gene mutation related to SCN or CyN and/or suffering from SCN or CyN wherein the ELANE gene mutation is selected from the group consisting of the ELANE mutants set forth in Table 1; [0135] b) introducing to the cells of step (a) a composition comprising: [0136] a CRISPR nuclease or a sequence encoding the CRISPR nuclease; and [0137] a first RNA molecule comprising a guide sequence portion having 17-30 nucleotides, wherein a complex of the CRISPR nuclease and the first RNA molecule affects a double strand break in the mutant allele of the ELANE gene in one or more cells, [0138] optionally, introducing to the cells a second RNA molecule comprising a guide sequence portion capable of complexing with a CRISPR nuclease, wherein the complex of the second RNA molecule and CRISPR nuclease affects a second double strand break in the ELANE gene in the one or more cells so as to inactivate the mutant allele of the ELANE gene in one or more cells thereby obtaining modified cells; optionally; [0139] c) culture expanding the cells of step (b) wherein the modified cells are capable of engraftment and giving rise to progeny cells after engraftment; and [0140] d) administering to the subject the cells of step (b) or step (c) for treating the SCN or CyN in the subject.

[0141] Embodiments of the present invention provide for a method of treating a subject afflicted with SCN or CyN, comprising administration of a therapeutically effective amount of the modified cells, compositions, or the compositions prepared by the methods of the instant invention

[0142] Embodiments of the present invention provide for a method for treating SCN or CyN in a subject with an ELANE gene mutation relating to SCN or CyN in need thereof wherein the ELANE gene mutation is selected from the group consisting of the ELANE mutants set forth in Table 1, the method comprising: [0143] a) isolating HSPCs from cells obtained from the subject; [0144] b) introducing to the cells of step (a) a composition comprising: [0145] a CRISPR nuclease or a sequence encoding the CRISPR nuclease; and [0146] a first RNA molecule comprising a guide sequence portion having 17-30 nucleotides, wherein a complex of the CRISPR nuclease and the first RNA molecule affects a double strand break in the mutant allele of the ELANE gene in one or more cells, [0147] optionally, introducing to the cells a second RNA molecule comprising a guide sequence portion capable of complexing with a CRISPR nuclease, wherein the complex of the second RNA molecule and CRISPR nuclease affects a second double strand break in the ELANE gene in the one or more cells so as to inactivate the mutant allele of the ELANE gene in one or more cells thereby obtaining modified cells; optionally; [0148] c) culture expanding the cells of step (b) wherein the modified cells are capable of engraftment and giving rise to progeny cells after engraftment; and [0149] d) administering to the subject the cells of step (b) or step (c) [0150] thereby treating the SCN or CyN in the subject.

[0151] Embodiments of the present invention provide for a method for treating SCN or CyN in a

subject with an ELANE gene mutation relating to SCN or CYN in need thereof wherein the ELANE gene mutation is selected from the group consisting of the ELANE mutants set forth in Table 1, the method comprising [0152] administering to the subject autologous modified cells or progeny of autologous modified cells, wherein the autologous modified cells are modified so as to have a double strand break in the mutant allele of the ELANE gene, [0153] wherein said double strand break results from introduction to the cells of a composition comprising a CRISPR nuclease or a sequence encoding the CRISPR nuclease and a first RNA molecule wherein a complex of the CRISPR nuclease and the first RNA molecule affects a double strand break in the mutant allele of the ELANE gene so as to inactivate the mutant allele of the ELANE gene in the cell, [0154] thereby treating the SCN or CyN in the subject.

[0155] Embodiments of the present invention provide for a method of selecting a subject for treatment from a pool of subjects diagnosed with SCN or CyN, comprising the steps of: [0156] a) obtaining cells from each subject in the pool of subjects; [0157] b) screening each subject's cells for an ELANE gene mutation related to SCN or CyN, and selecting only subjects with an ELANE gene mutation related to SCN or CyN; [0158] c) screening by sequencing the cells of the subjects selected in step (b) for a mutant allele of the ELANE gene selected from the group consisting of the ELANE mutations set forth in Table 1, [0159] d) selecting for treatment only subjects with cells heterozygous for an ELANE mutation set forth in Table 1. [0160] e) obtaining HSPC cells from the bone marrow of the subject either by aspiration or by mobilization and apheresis of peripheral blood; and [0161] f) introducing to the HSPC cells of step (e): [0162] one or more CRISPR nucleases or sequences encoding the one or more CRISPR nucleases [0163] a first RNA molecule comprising a guide sequence portion having 17-30 contiguous nucleotides containing nucleotides in the sequence set forth in any one of SEQ ID NOs: 1-31117 targeting the mutant allele of the ELANE gene, and [0164] a second RNA molecule comprising a guide sequence portion targeting a sequence in intron 3, intron 4 or 3' UTR of the ELANE gene, wherein a complex of the first RNA molecule and a CRISPR nuclease affects a first double strand break in the mutant allele of the ELANE gene in one or more of the HSPC cells and a complex of the second RNA molecule and a CRISPR nuclease affect a second double strand break in intron 3, intron 4, or 3' UTR of both alleles of the ELANE gene in the one or more HSPC cells in which the complex of the first RNA molecule and the CRISPR nuclease affected a first double strand break, thereby obtaining modified cells; [0165] g) administering to the subject the modified cells of step (f), [0166] thereby treating SCN or CyN in the subject.

[0167] Embodiments of the present invention provide an RNA molecule comprising a guide sequence portion having 17-30 contiguous nucleotides containing nucleotides in the sequence set forth in any one of SEQ ID NOs: 1-31117.

[0168] Embodiments of the present invention further comprise a second RNA molecule comprising a guide sequence portion.

[0169] In embodiments of the present invention, the second RNA molecule targets a non-coding region of the ELANE gene.

[0170] In embodiments of the present invention, the nucleotide sequence of the guide sequence portion of the second RNA molecule is a different nucleotide sequence from the sequence of the guide sequence portion of the first RNA molecule.

[0171] In embodiments of the present invention, the first RNA molecule further comprise a portion having a sequence which binds to a CRISPR nuclease. In embodiments of the present invention, the second RNA molecule further comprise a portion having a sequence which binds to a CRISPR nuclease.

[0172] In an embodiment of the present invention: [0173] a. the CRISPR nuclease has at least 100%, 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, 90%, 85%, 80% identity to SEQ ID NO: 31123, and the sequence which binds to a CRISPR nuclease comprises a sequence selected from SEQ ID NOs: 31151 to 31160; [0174] b. the CRISPR nuclease has at least 100%, 99%, 98%,

97%, 96%, 95%, 94%, 93%, 92%, 91%, 90%, 85%, 80% identity to SEQ ID NO: 31124, and the sequence which binds to a CRISPR nuclease comprises a sequence selected from SEQ ID NOs: 31161 to 31170; [0175] c. the CRISPR nuclease has at least 100%, 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, 90%, 85%, 80% identity to SEQ ID NO: 31125, and the sequence which binds to a CRISPR nuclease comprises a sequence selected from SEQ ID NOs: 31171 to 31181; or [0176] d. the CRISPR nuclease has at least 100%, 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, 90%, 85%, 80% identity to SEQ ID NO: 31126, and the sequence which binds to a CRISPR nuclease comprises a sequence selected from SEQ ID NOs: 31182 to 31189 and GGAUUAUCC. [0177] In embodiments of the present invention, the sequence which binds to a CRISPR nuclease is a tracrRNA sequence.

[0178] In embodiments of the present invention, the first RNA molecule further comprises a portion having a tracr mate sequence. In embodiments of the present invention, the second RNA molecule further comprises a portion having a tracr mate sequence.

[0179] In embodiments of the present invention, the first RNA molecule further comprises one or more linker portions. In embodiments of the present invention, the second RNA molecule further comprises one or more linker portions.

[0180] In embodiments of the present invention, the first RNA molecule is up to 300 nucleotides in length. In embodiments of the present invention, the second RNA molecule is up to 300 nucleotides in length.

[0181] In embodiments of the present invention, the composition further comprises one or more CRISPR nucleases or sequences encoding the one or more CRISPR nucleases. In embodiments of the present invention, the composition further comprises one or more tracrRNA molecules or sequences encoding the one or more tracrRNA molecules.

[0182] Embodiments of the present invention provide a method for inactivating in a cell a mutant ELANE allele, the method comprising delivering to the cell the RNA molecules or compositions of the present invention.

[0183] In embodiments of the present invention, the one or more CRISPR nuclease or sequences encoding the one or more CRISPR nucleases and the RNA molecule or RNA molecules are delivered to the subject and/or cells substantially at the same time or at different times.

[0184] In embodiments of the present invention, the tracrRNA molecules or sequences encoding the one or more tracrRNA molecules and the RNA molecule or RNA molecules are delivered to the subject and/or cells substantially at the same time or at different times.

[0185] In embodiments of the present invention, the method comprises removing an exon containing a disease-causing mutation from a mutant allele, wherein the first RNA molecule or the first and the second RNA molecules target regions flanking an entire exon or a portion of the exon.

[0186] In embodiments of the present invention, the method comprises removing multiple exons, the entire open reading frame of a gene, or removing the entire gene.

[0187] In embodiments of the present invention, the first RNA molecule or the first and the second RNA molecules target an alternative splicing signal sequence between an exon and an intron of a mutant allele.

[0188] In embodiments of the present invention, the second RNA molecule targets a sequence present in both a mutant allele and a functional allele.

[0189] In embodiments of the present invention, the second RNA molecule targets an intron.

[0190] In embodiments of the present invention, the method results in subjecting the mutant allele to insertion or deletion by an error prone non-homologous end joining (NHEJ) mechanism, generating a frameshift in the mutant allele's sequence.

[0191] In embodiments of the present invention, the frameshift results in inactivation or knockout of the mutant allele.

[0192] In embodiments of the present invention, the frameshift creates an early stop codon in the mutant allele or the frameshift results in nonsense-mediated mRNA decay of the transcript of the

mutant allele.

[0193] In embodiments of the present invention, inactivating or treating results in a truncated protein encoded by the mutant allele and a functional protein encoded by the functional allele.

[0194] Embodiments of the present invention provide use of the RNA molecules, the compositions, or the composition prepared by the method of the present invention for inactivating in a cell a mutant ELANE allele.

[0195] Embodiments of the present invention provide a medicament comprising the RNA molecules, compositions, or the compositions prepared by the methods of the instant invention for use in inactivating in a cell a mutant ELANE allele, wherein the medicament is administered by delivering to the cell the RNA molecules, compositions, or the compositions prepared by the methods of the instant invention.

[0196] Embodiments of the present invention provide for use of the methods, the modified cells, the compositions, or the compositions prepared by the methods, or the RNA molecules of the instant invention for treating ameliorating or preventing SCN or CyN in to a subject having or at risk of having SCN or CyN.

[0197] Embodiments of the present invention provide for a medicament comprising the RNA molecules, compositions, compositions prepared by the methods of the instant invention, or the modified cells of the instant invention, for use in treating ameliorating or preventing SCN or CyN, wherein the medicament is administered by delivering to a subject having or at risk of having SCN or CyN the RNA molecules, compositions, compositions prepared by the methods of the instant invention, or the modified cells of the instant invention.

[0198] Embodiments of the present invention provide for a kit for inactivating a mutant ELANE allele in a cell, comprising the RNA molecules of the instant invention, a CRISPR nuclease or sequence encoding the CRISPR nuclease, and/or a tracrRNA molecule or sequence encoding the tracrRNA molecule; and instructions for delivering the RNA molecule; CRISPR nuclease or sequence encoding the CRISPR nuclease, and/or the tracrRNA or sequence encoding the tracrRNA molecule to the cell to inactivate the mutant ELANE allele in the cell.

[0199] Embodiments of the present invention provide for a kit for treating SCN or CyN in a subject, comprising the RNA molecules of the instant invention, a CRISPR nuclease or sequence encoding the CRISPR nuclease, and/or a tracrRNA molecule or sequence encoding the tracrRNA molecule; and instructions for delivering the RNA molecule; CRISPR nuclease or sequence encoding the CRISPR nuclease, and/or the tracrRNA or sequence encoding the tracrRNA molecule to a subject having or at risk of having SCN or CyN so as to treat the SCN or CyN.

[0200] Embodiments of the present invention provide for a kit for inactivating a mutant ELANE allele in a cell, comprising the compositions, the composition prepared by the methods of the instant invention, or the modified cells of the instant invention, and instructions for delivering the composition to the cell so as to inactivate the ELANE gene in the cell.

[0201] Embodiments of the present invention provide for a kit for treating SCN or CyN in a subject, comprising the composition, the compositions prepared by the methods of the instant invention, or the modified cells of the instant invention, and instructions for delivering the compositions, the compositions prepared by the methods of the instant invention, or the modified cells of the instant invention, to a subject having or at risk of having SCN or CyN so as to treat SCN or CyN.

#### Definitions

[0202] Unless otherwise defined, all technical and/or scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the invention pertains. Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of embodiments of the invention, exemplary methods and/or materials are described below. In case of conflict, the patent specification, including definitions, will control. In addition, the materials, methods, and examples are illustrative only and are not

intended to be necessarily limiting.

[0203] It should be understood that the terms “a” and “an” as used above and elsewhere herein refer to “one or more” of the enumerated components. It will be clear to one of ordinary skill in the art that the use of the singular includes the plural unless specifically stated otherwise. Therefore, the terms “a,” “an” and “at least one” are used interchangeably in this application.

[0204] For purposes of better understanding the present teachings and in no way limiting the scope of the teachings, unless otherwise indicated, all numbers expressing quantities, percentages or proportions, and other numerical values used in the specification and claims, are to be understood as being modified in all instances by the term “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the following specification and attached claims are approximations that may vary depending upon the desired properties sought to be obtained. At the very least, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques.

[0205] Unless otherwise stated, adjectives such as “substantially” and “about” modifying a condition or relationship characteristic of a feature or features of an embodiment of the invention, are understood to mean that the condition or characteristic is defined to within tolerances that are acceptable for operation of the embodiment for an application for which it is intended. Unless otherwise indicated, the word “or” in the specification and claims is considered to be the inclusive “or” rather than the exclusive or, and indicates at least one of, or any combination of items it conjoins.

[0206] In the description and claims of the present application, each of the verbs, “comprise,” “include” and “have” and conjugates thereof, are used to indicate that the object or objects of the verb are not necessarily a complete listing of components, elements or parts of the subject or subjects of the verb. Other terms as used herein are meant to be defined by their well-known meanings in the art.

[0207] The “guide sequence portion” of an RNA molecule refers to a nucleotide sequence that is capable of hybridizing to a specific target DNA sequence, e.g., the guide sequence portion has a nucleotide sequence which is fully complementary to the DNA sequence being targeted along the length of the guide sequence portion. In some embodiments, the guide sequence portion is 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 nucleotides in length, or approximately 17-30, 17-29, 17-28, 17-27, 17-26, 17-25, 17-24, 18-22, 19-22, 18-20, 17-20, or 17-22 nucleotides in length. The entire length of the guide sequence portion is fully complementary to the DNA sequence being targeted along the length of the guide sequence portion. The guide sequence portion may be part of an RNA molecule that can form a complex with a CRISPR nuclease with the guide sequence portion serving as the DNA targeting portion of the CRISPR complex. When the DNA molecule having the guide sequence portion is present contemporaneously with the CRISPR molecule the RNA molecule is capable of targeting the CRISPR nuclease to the specific target DNA sequence. Each possibility represents a separate embodiment. An RNA molecule can be custom designed to target any desired sequence.

[0208] The term “targets” as used herein, refers to the guide sequence portion of the RNA molecule's preferential hybridization to a nucleic acid having a targeted nucleotide sequence. It is understood that the term “targets” encompasses variable hybridization efficiencies, such that there is preferential targeting of the nucleic acid having the targeted nucleotide sequence, but unintentional off-target hybridization in addition to on-target hybridization might also occur. It is understood that where an RNA molecule targets a sequence, a complex of the RNA molecule and a CRISPR nuclease molecule targets the sequence for nuclease activity.

[0209] As used herein, the term “targeting sequence” or “targeting molecule” refers a nucleotide sequence or molecule comprising a nucleotide sequence that is capable of hybridizing to a specific target sequence, e.g., the targeting sequence has a nucleotide sequence which is at least partially complementary to the sequence being targeted along the length of the targeting sequence. The

targeting sequence or targeting molecule may be part of an RNA molecule that can form a complex with a CRISPR nuclease with the targeting sequence serving as the targeting portion of the CRISPR complex. When the molecule having the targeting sequence is present contemporaneously with the CRISPR molecule the RNA molecule is capable of targeting the CRISPR nuclease to the specific target sequence.

[0210] In the context targeting a DNA sequence that is present in a plurality of cells, it is understood that the targeting encompasses hybridization of the guide sequence portion of the RNA molecule with the sequence in one or more of the cells, and also encompasses hybridization of the RNA molecule with the target sequence in fewer than all of the cells in the plurality of cells.

Accordingly, it is understood that where an RNA molecule targets a sequence in a plurality of cells, a complex of the RNA molecule and a CRISPR nuclease is understood to hybridize with the target sequence in one or more of the cells, and also may hybridize with the target sequence in fewer than all of the cells. Accordingly, it is understood that the complex of the RNA molecule and the CRISPR nuclease introduces a double strand break in relation to hybridization with the target sequence in one or more cells and may also introduce a double strand break in relation to hybridization with the target sequence in fewer than all of the cells. As used herein, the term “modified cells” refers to cells in which a double strand break is effected by a complex of an RNA molecule and the CRISPR nuclease as a result of hybridization with the target sequence, i.e. on-target hybridization.

[0211] The term “non-discriminatory” as used herein refers to a guide sequence portion of an RNA molecule that targets a specific DNA sequence that is common both a mutant and functional allele of a gene.

[0212] In embodiments of the present invention, RNA guide molecule may target the ELANE mutant allele based on the mutation associated with severe congenital neutropenia (SCN) or cyclic neutropenia (CyN).

[0213] In embodiments of the present invention, an RNA molecule comprises a guide sequence portion having 17-30 contiguous nucleotides containing nucleotides in the sequence set forth in any one of SEQ ID NOs: 1-31117.

[0214] It is understood that in any of the embodiments of the present invention the guide sequence portion of an RNA molecule may comprise 17-22 contiguous nucleotides set forth in any single sequence of SEQ ID NOs: 1-31117, or in any single sequence from the above groups of sequences.

[0215] The RNA molecule and or the guide sequence portion of the RNA molecule may contain modified nucleotides. Exemplary modifications to nucleotides or polynucleotides may be synthetic and encompass polynucleotides which contain nucleotides comprising bases other than the naturally occurring adenine, cytosine, thymine, uracil, or guanine bases. Modifications to polynucleotides include polynucleotides which contain synthetic, non-naturally occurring nucleosides e.g., locked nucleic acids. Modifications to polynucleotides may be utilized to increase or decrease stability of an RNA. An example of a modified polynucleotide is an mRNA containing 1-methyl pseudo-uridine. For examples of modified polynucleotides and their uses, see U.S. Pat. No. 8,278,036, PCT International Publication No. WO/2015/006747, and Weissman and Kariko (2015), hereby incorporated by reference.

[0216] As used herein, “contiguous nucleotides” set forth in a SEQ ID NO refers to nucleotides in a sequence of nucleotides in the order set forth in the SEQ ID NO without any intervening nucleotides.

[0217] In embodiments of the present invention, the guide sequence portion may be 20 nucleotides in length and consists of 20 nucleotides in the sequences set forth in any one of SEQ ID NOs: 1-31117. In embodiments of the present invention, the guide sequence portion may be less than 20 nucleotides in length. For example, in embodiments of the present invention the guide sequence portion may be 17, 18, or 19 nucleotides in length. In such embodiments the guide sequence portion may consist of 17, 18, or 19 nucleotides, respectively, in the sequences set forth in any one

of SEQ ID NOs: 1-31117. For example, a guide sequence portion having 17 nucleotides in the sequence of 17 contiguous nucleotides set forth in SEQ ID NO: 31118 may consist of any one of the following nucleotide sequences (nucleotides excluded from the contiguous sequence are marked in strike-through):

TABLE-US-00001 (SEQ ID NO: 31118) AAAAAAACCAGACGCCGGCC 17  
nucleotide guide sequence 1: (SEQ ID NO: 31119)  custom-character  
AAAACCAGACGCCGGCC 17 nucleotide guide sequence 2: (SEQ ID NO:  
31120)  custom-character AAAAACCAGACGCCGGC  custom-character 17 nucleotide  
guide sequence 3: (SEQ ID NO: 31121)  custom-character AAAAAACCAGACGCCGG  
 custom-character 17 nucleotide guide sequence 4: (SEQ ID NO: 31122)  
AAAAAAACCAGACGCCG  custom-character

[0218] In embodiments of the present invention, the guide sequence portion may be greater than 22 nucleotides in length. For example, in embodiments of the present invention the guide sequence portion may be 23, 24, 25, 26, 27, 28, 29, or 30 nucleotides in length. In such embodiments the guide sequence portion comprises 20, 21, or 22 nucleotides in the sequences set forth in any one of SEQ ID NOs: 1-31117 and additional nucleotides fully complementary to a nucleotide or sequence of nucleotides adjacent to the 3' end of the target sequence, 5' end of the target sequence, or both.

[0219] In embodiments of the present invention, a CRISPR nuclease and an RNA molecule comprising a guide sequence portion form a CRISPR complex that binds to a target DNA sequence to effect cleavage of the target DNA sequence. CRISPR nucleases, e.g. Cpf1, may form a CRISPR complex comprising a CRISPR nuclease and RNA molecule without a further tracrRNA molecule. Alternatively, CRISPR nucleases, e.g. Cas9, may form a CRISPR complex between the CRISPR nuclease, an RNA molecule, and a tracrRNA molecule.

[0220] In embodiments of the present invention, the RNA molecule may further comprise the sequence of a tracrRNA molecule. Such embodiments may be designed as a synthetic fusion of the guide portion of the RNA molecule and the trans-activating crRNA (tracrRNA). (See Jinek (2012) Science). Embodiments of the present invention may also form CRISPR complexes utilizing a separate tracrRNA molecule and a separate RNA molecule comprising a guide sequence portion. In such embodiments the tracrRNA molecule may hybridize with the RNA molecule via base pairing and may be advantageous in certain applications of the invention described herein.

[0221] The term “tracr mate sequence” refers to a sequence sufficiently complementary to a tracrRNA molecule so as to hybridize to the tracrRNA via basepairing and promote the formation of a CRISPR complex. (See U.S. Pat. No. 8,906,616). In embodiments of the present invention, the RNA molecule may further comprise a portion having a tracr mate sequence.

[0222] According to embodiments of the present invention, an RNA molecule may be up to 300, 290, 280, 270, 260, 250, 240, 230, 220, 210, 200, 190, 180, 170, 160, 150, 140, 130, 120, 110, or 100 nucleotides in length. Each possibility represents a separate embodiment. In embodiments of the present invention, the RNA molecule may be 17 up to 300 nucleotides in length, 100 up to 300 nucleotides in length, 150 up to 300 nucleotides in length, 200 up to 300 nucleotides in length, 100 to 200 nucleotides in length, or 150 up to 250 nucleotides in length. Each possibility represents a separate embodiment.

[0223] A “gene,” for the purposes of the present disclosure, includes a DNA region encoding a gene product, as well as all DNA regions which regulate the production of the gene product, whether or not such regulatory sequences are adjacent to coding and/or transcribed sequences. Accordingly, a gene includes, but is not necessarily limited to, promoter sequences, terminators, translational regulatory sequences such as ribosome binding sites and internal ribosome entry sites, enhancers, silencers, insulators, boundary elements, replication origins, matrix attachment sites and locus control regions.

[0224] “Eukaryotic” cells include, but are not limited to, fungal cells (such as yeast), plant cells, animal cells, mammalian cells and human cells.



[0225] As used herein, the term HSPC refers to both hematopoietic stem cells and hematopoietic stem progenitor cells. Non-limiting examples of stem cells include a bone marrow cell, a myeloid progenitor cell, a multipotent progenitor cell, a lineage restricted progenitor cell.

[0226] As used herein, “progenitor cell” refers to a lineage cell that is derived from stem cell and retains mitotic capacity and multipotency (e.g., can differentiate or develop into more than one but not all types of mature lineage of cell). As used herein “hematopoiesis” or “hemopoiesis” refers to the formation and development of various types of blood cells (e.g., red blood cells, megakaryocytes, myeloid cells (e.g., monocytes, macrophages and neutrophil), and lymphocytes) and other formed elements in the body (e.g., in the bone marrow).

[0227] The term “nuclease” as used herein refers to an enzyme capable of cleaving the phosphodiester bonds between the nucleotide subunits of nucleic acid. A nuclease may be isolated or derived from a natural source. The natural source may be any living organism. Alternatively, a nuclease may be a modified or a synthetic protein which retains the phosphodiester bond cleaving activity. Gene modification can be achieved using a nuclease, for example a CRISPR nuclease.

[0228] According to embodiments of the present invention, the first RNA molecule or the first and the second RNA molecules targets at least a portion of the promoter and/or the start codon and/or a portion of the UTR of the mutant allele of the ELANE gene.

[0229] According to embodiments of the present invention, there is provided a method comprising removing an exon containing a disease-causing mutation from a mutant allele, wherein the first RNA molecule or the first and the second RNA molecules target regions flanking an entire exon or a portion of the exon.

[0230] According to embodiments of the present invention, there is provided a method comprising removing multiple exons, the entire open reading frame of a gene, or removing the entire gene.

[0231] According to embodiments of the present invention, the first RNA molecule targets a mutant allele of the ELANE gene, and the second RNA molecule targets a sequence in an intron present in both the mutant and functional allele of the ELANE gene.

[0232] According to embodiments of the present invention, the second RNA molecule targets a sequence present in both a mutant allele and a functional allele of the ELANE gene.

[0233] According to embodiments of the present invention, the second RNA molecule targets an intron.

[0234] According to embodiments of the present invention, there is provided a method comprising subjecting the mutant allele to insertion or deletion by an error prone non-homologous end joining (NHEJ) mechanism, generating a frameshift in the mutant allele's sequence.

[0235] According to embodiments of the present invention, the frameshift results in inactivation or knockout of the mutant allele.

[0236] According to embodiments of the present invention, the frameshift creates an early stop codon in the mutant allele.

[0237] According to embodiments of the present invention, the frameshift results in nonsense-mediated mRNA decay of the transcript of the mutant allele.

[0238] According to embodiments of the present invention, the inactivating or treating results in a truncated protein encoded by the mutant allele and a functional protein encoded by the functional allele.

[0239] The compositions and methods of the present disclosure may be utilized for treating, preventing, ameliorating, or slowing progression of SCN or CyN.

[0240] In some embodiments, the method of deactivating a mutant allele comprises an exon skipping step comprising removing an exon containing a disease-causing mutation from the mutant allele. Removing an exon containing a disease-causing mutation in the mutant allele requires two RNA molecules which target regions flanking the entire exon or a portion of the exon. Removal of an exon containing the disease-causing mutation may be designed to eliminate the disease-causing action of the protein while allowing for expression of the remaining protein product which retains

some or all of the wild-type activity. As an alternative to single exon skipping, multiple exons, the entire open reading frame or the entire gene can be excised using two RNA molecules flanking the region desired to be excised.

[0241] In some embodiments, the method of deactivating a mutant allele comprises delivering two RNA molecules to a cell, wherein one RNA molecule targets a first mutant allele of an ELANE gene and wherein the other RNA molecule targets a second mutant allele of the ELANE gene, or the second RNA molecule targets a sequence in an intron present in both the mutant or functional allele.

[0242] In some embodiments, an RNA molecule is used to target a CRISPR nuclease to an alternative splicing signal sequence between an exon and an intron of a mutant allele, thereby destroying the alternative splicing signal sequence in the mutant allele.

[0243] Any one of, or combination of, the above-mentioned strategies for deactivating a mutant allele may be used in the context of the invention.

[0244] Additional strategies may be used to deactivate a mutant allele. For example, in embodiments of the present invention, an RNA molecule is used to direct a CRISPR nuclease to an exon or a splice site of a mutant allele in order to create a double-stranded break (DSB), leading to insertion or deletion of nucleotides by an error-prone non-homologous end-joining (NHEJ) mechanism and formation of a frameshift mutation in the mutant allele. The frameshift mutation may result in: (1) inactivation or knockout of the mutant allele by generation of an early stop codon in the mutant allele, resulting in generation of a truncated protein; or (2) nonsense mediated mRNA decay of the transcript of the mutant allele. In further embodiments, one RNA molecule is used to direct a CRISPR nuclease to a promotor of a mutant allele.

[0245] In some embodiments, the method of deactivating a mutant allele further comprises enhancing activity of the functional protein such as by providing a protein/peptide, a nucleic acid encoding a protein/peptide, or a small molecule such as a chemical compound, capable of activating/enhancing activity of the functional protein.

[0246] According to some embodiments, the present disclosure provides an RNA molecule which binds to/associates with and/or directs the RNA guided DNA nuclease e.g., CRISPR nuclease to a mutant of the ELANE gene.

[0247] In some embodiments, the method comprises the steps of: contacting a mutant allele of a gene of interest with an allele-specific RNA molecule and a CRISPR nuclease e.g., a Cas9 protein, wherein the allele-specific RNA molecule and the CRISPR nuclease e.g., Cas9 associate with a nucleotide sequence of the mutant allele of the gene of interest which differs by at least one nucleotide from a nucleotide sequence of a functional allele of the gene of interest, thereby modifying or knocking-out the mutant allele.

[0248] In some embodiments, the allele-specific RNA molecule and a CRISPR nuclease is introduced to a cell encoding the gene of interest. In some embodiments, the cell encoding the gene of interest is in a mammalian subject. In some embodiments, the cell encoding the gene of interest is in a plant.

[0249] In some embodiments, the cleaved mutant allele is further subjected to insertion or deletion (indel) by an error prone non-homologous end joining (NHEJ) mechanism, generating a frameshift in the mutant allele's sequence. In some embodiments, the generated frameshift results in inactivation or knockout of the mutant allele. In some embodiments, the generated frameshift creates an early stop codon in the mutant allele and results in generation of a truncated protein. In such embodiments, the method results in the generation of a truncated protein encoded by the mutant allele and a functional protein encoded by the functional allele. In some embodiments, a frameshift generated in a mutant allele using the methods of the invention results in nonsense-mediated mRNA decay of the transcript of the mutant allele.

[0250] A skilled artisan will appreciate that in each of the embodiments of the present invention, individually, each of the RNA molecules of the present invention are capable of complexing with a

nuclease, e.g. a CRISPR nuclease, such as to associate with a target genomic DNA sequence of interest next to a protospacer adjacent motif (PAM). The nuclease then mediates cleavage of target DNA to create a double-stranded break within the protospacer. Accordingly, in embodiments of the present invention, the guide sequences and RNA molecules of the present invention may target a location 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, or 28 nucleotides upstream or downstream from a PAM site.

[0251] Therefore, in embodiments of the present invention, an RNA molecule of the present invention in complex with a nuclease, e.g., a CRISPR nuclease, may affect a double strand break in an allele of a gene 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28 upstream or downstream from a target site.

[0252] Where the target site (e.g. a pathogenic mutation) is within the PAM site, it is understood that the RNA molecule may be designed to target a sequence 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, or 28 nucleotides upstream or downstream from the PAM site. Accordingly, an RNA molecule is designed to target a nuclease, e.g. a CRISPR nuclease, only to an allele containing the pathogenic mutation.

[0253] In embodiments of the present invention, an RNA molecule is designed to target a mutant allele in the ELANE gene, wherein the RNA molecule and/or the complex of the RNA molecule and a CRISPR nuclease targets the nucleotide base present in the mutant allele of the ELANE gene

[0254] In embodiments of the present invention, the RNA molecules, compositions, methods, cells, kits, or medicaments are utilized for treating a subject having a disease phenotype resulting from the heterozygote ELANE gene. In embodiments of the present invention, the disease is SCN or CyN. In such embodiments, the method results in improvement, amelioration or prevention of the disease phenotype.

[0255] In embodiments of the present invention, the RNA molecules, compositions, methods, cells, kits, or medicaments of the present invention are utilized in combination with a second therapy for SCN or CyN to treat the subject. In embodiments of the present invention, the RNA molecules, compositions, methods, kits, or medicaments of the present invention are administered prior to administration of the second therapy, during administration of the second therapy, and/or after administration of the second therapy.

[0256] In embodiments of the present invention, the RNA molecules, compositions, methods, cells, kits, or medicaments of the present invention are administered in combination with Granulocyte colony-stimulating factor (G-CSF) therapy.

[0257] Embodiments referred to above refer to a CRISPR nuclease, RNA molecule(s), and tracrRNA being effective in a subject or cells at the same time. The CRISPR, RNA molecule(s), and tracrRNA can be delivered substantially at the same time or can be delivered at different times but have effect at the same time. For example, this includes delivering the CRISPR nuclease to the subject or cells before the RNA molecule and/or tracr RNA is substantially extant in the subject or cells.

[0258] According to embodiments of the present invention, there is provided a method for inactivating in a cell a mutant allele of the ELANE gene, the method comprising the steps of:

[0259] a) selecting a cell with an ELANE gene mutation associated with SCN or CyN wherein the ELANE gene mutation is selected from the group consisting of the ELANE mutants set forth in

Table 1; [0260] b) introducing to the cell a composition comprising:

[0261] a CRISPR nuclease or sequence encoding the CRISPR nuclease, and [0262] a first RNA molecule comprising a guide sequence portion having 17-30 nucleotides, [0263] wherein a complex of the CRISPR nuclease and

the first RNA molecule affects a double strand break in the mutant allele of the ELANE gene and not in the functional allele of the ELANE gene in the cell; [0264] thereby inactivating the mutant allele of the ELANE gene in the cell.

[0265] According to embodiments of the present invention, there is provided a method for inactivating in a cell a mutant allele of the ELANE gene, the method comprising the steps of:

[0266] a) selecting a cell with an ELANE gene mutation associated with SCN or CyN wherein the ELANE gene mutation is selected from the group consisting of the ELANE mutants set forth in Table 1; [0267] b) introducing to the cell a composition comprising: [0268] a CRISPR nuclease or sequence encoding the CRISPR nuclease, and [0269] a first RNA molecule comprising a guide sequence portion having 17-30 nucleotides, [0270] wherein a complex of the CRISPR nuclease and the first RNA molecule affects a double strand break in the mutant allele of the ELANE gene and not in the functional allele of the ELANE gene in the cell; [0271] and wherein the method further comprises introduction of a second RNA molecule comprising a guide sequence portion capable of complexing with a CRISPR nuclease, wherein the complex of the second RNA molecule and the CRISPR nuclease affects a second double strand break in the ELANE gene; thereby inactivating the mutant allele of the ELANE gene in the cell.

[0272] According to embodiments of the present invention, there is provided a method for inactivating in a cell a mutant allele of the ELANE gene having a mutation associated with SCN or CyN wherein the mutant allele is selected from the group consisting of the ELANE mutants set forth in Table 1, the method comprising [0273] introducing to the cell a composition comprising: [0274] a CRISPR nuclease or sequence encoding the CRISPR nuclease, and [0275] a first RNA molecule comprising a guide sequence portion having 17-30 nucleotides, [0276] wherein a complex of the CRISPR nuclease and the first RNA molecule affects a double strand break in the mutant allele of the ELANE gene; [0277] thereby inactivating the mutant allele of the ELANE gene in the cell.

[0278] According to embodiments of the present invention, there is provided a method for inactivating in a cell a mutant allele of the ELANE gene associated with SCN or CyN wherein the mutant allele is selected from the group consisting of the ELANE mutants set forth in Table 1, the method comprising: [0279] introducing to the cell a composition comprising: [0280] a CRISPR nuclease or sequence encoding the CRISPR nuclease, and [0281] a first RNA molecule comprising a guide sequence portion having 17-30 nucleotides, [0282] wherein a complex of the CRISPR nuclease and the first RNA molecule affects a double strand break in the mutant allele of the ELANE gene; [0283] and wherein the method further comprises introduction of a second RNA molecule comprising a guide sequence portion capable of complexing with a CRISPR nuclease, wherein the complex of the second RNA molecule and CRISPR nuclease affects a second double strand break in the ELANE gene; [0284] thereby inactivating the mutant allele of the ELANE gene in the cell.

[0285] In embodiments of the present invention, a complex of the CRISPR nuclease and the first RNA molecule affects a double strand break in the mutant allele of the ELANE gene and not in the functional allele of the ELANE gene in the cell.

[0286] In embodiments of the present invention, the cell comprises a second mutant allele selected from the group consisting of the ELANE mutants set forth in Table 1.

[0287] In embodiments of the present invention, a cell with an ELANE gene mutation associated with SCN or CyN may be from a subject with the ELANE gene mutation and/or afflicted with SCN or CyN. Accordingly, selecting a cell with an ELANE gene mutation may comprise selecting a subject with the ELANE gene mutation. In further embodiments of the present invention, selecting a cell may comprise selecting a cell from a subject with the ELANE gene mutation. In embodiments of the present invention, introducing the compositions of the subject invention to the cell may comprise introducing the compositions of the invention to the cell of a subject afflicted with the ELANE gene mutation.

[0288] Accordingly, in embodiments of the present invention, there is provided a method for inactivating in a cell a mutant allele of the ELANE gene of a subject, the method comprising the step of selecting a subject with an ELANE gene mutation resulting in SCN or CyN wherein the ELANE gene mutation is selected from the group consisting of the ELANE mutants set forth in Table 1.

[0289] Accordingly, embodiments of the present invention encompass the screening of subjects or cells for the ELANE gene. A person having ordinary skill in the art would readily understand methods of screening for mutations within the ELANE gene in the art, by way of non-limiting examples, e.g., sequencing-by-synthesis, Sanger sequencing, karyotyping, Fluorescence In situ Hybridization, and/or microarray testing. In embodiments of the present invention, mutations within the ELANE gene are screened by exon sequencing.

[0290] In embodiments of the present invention, the subject is or has been diagnosed with SCN or CyN by measuring the absolute neutrophil count (ANC) in peripheral blood. In embodiments of the present invention, SCN is or was diagnosed before the subject reaches the age 6 months. In embodiments of the present invention, CyN is or was diagnosed between the ages of 12 and 24 months, or after the age of 24 months. In embodiments of the present invention, SCN or Cyn is diagnosed by one or more of recurrent acute stomatologic disorders. In embodiments of the present invention, SCN or CyN is diagnosed by bone marrow examination, preferably the bone marrow examination is a cytogenetic bone marrow study. In embodiments of the present invention, SCN or CyN is diagnosed by one or more of: antineutrophil antibody assay, immunoglobulin assay (Ig GAM), lymphocyte immunophenotyping, pancreatic markers (serum trypsinogen and fecal elastase) and liposoluble vitamin levels (vitamins A, E and D). It is understood that any diagnostic method may be used with any other diagnostic method.

[0291] In embodiments of the present invention, a subject diagnosed with SCN or CyN is screened by Exon sequencing to identify an ELANE pathogenic mutation in the ELANE gene. In further embodiments the subject is screened by Sanger sequencing to confirm heterozygosity of at least one of the ELANE gene mutations selected from the group consisting of the ELANE mutants set forth in Table 1. In embodiments of the present invention, the nucleotide of the mutant allele of the ELANE gene is determined using BAC bio. In embodiments of the present invention, appropriate guides are selected according to Tables 2A-2C. In embodiments of the present invention, the guides selected are introduced to cells, e.g. PBMCs, obtained from the subject and reduction in the pathogenic ELANE mutation in the cells is measured by, e.g. Next Generation Sequencing.

[0292] It is understood that the CRISPR/Cas9 gene editing system enables targeting the nuclease to a target site in a sequence specific manner to address disease-causing mutations. Hematopoietic stem and progenitor cells (HSPCs) have therapeutic potential because of their ability to both self-renew and differentiate (Yu, Natanson, and Dunbar 2016). Accordingly, embodiments of the present invention apply genome editing to HSPCs.

[0293] In embodiments of the present invention, an autologous therapy and utilizes autologous CD34+ hematopoietic stem cells from patients diagnosed with SCN or CyN which are edited with CRISPR/Cas9. In embodiments of the present invention, CD34+ cells are isolated from bone marrow or peripheral blood mononucleated cells (PBMCs) following patient apheresis.

[0294] In the case of dominant negative (or compound heterozygous) indications, such as SCN or CyN, the strategy is to edit the mutant allele and avoid cleavage in the non-mutant allele or other off targets by targeting the mutant allele sequence.

[0295] Embodiments of the present invention may include the following steps: [0296] selection of a patient diagnosed with SCN or CyN identified as having a mutant allele of the ELANE gene selected from the group consisting of the ELANE mutants set forth in Table 1; [0297] selection of a therapeutic strategy based on the identified mutant allele of the candidate patient; [0298] obtaining HSPC cells from the bone marrow of the subject either by aspiration or by mobilization and apheresis of peripheral blood, optionally, the HSPC cells are processed (e.g., enriched, stimulated, both); [0299] introducing into the HSPC cells (e.g., by ex vivo electroporation) a composition comprising: [0300] a CRISPR nuclease or a sequence encoding the same (e.g., mRNA), [0301] a discriminatory RNA molecule that targets the mutant allele, and [0302] a non-discriminatory RNA molecule targeting a sequence in intron 3, intron 4 or 3' UTR, which is common to both the mutant allele and the other allele, [0303] thereby editing the HSPC cells to knockout expression of mutant

ELANE allele; and [0304] introducing the edited HSPC to the candidate patient.

[0305] In embodiments of the present invention, CD34<sup>+</sup> cells may be isolated from bone marrow or peripheral blood mononucleated cells (PBMCs) following patient apheresis. Bone marrow or PBMCs may be collected from the patient by apheresis following HSPC mobilization. In embodiments of the invention the apheresis product may be washed to remove platelets and a CD34<sup>+</sup> cell population may be enriched via purification using, e.g. a CliniMACS system (Miltenyi Biotec). In embodiments of the present invention, the selected cells may be prestimulated ex vivo, e.g. with a mixture of recombinant human cytokines. In embodiments of the present invention, the cells may undergo electroporation. In embodiments of the present invention, prior to electroporation, stimulated cells (e.g. CD34<sup>+</sup> cells), the CRISPR nuclease mRNA and gRNA may be preincubated under defined conditions. In embodiments of the present invention, the cells are electroporated ex vivo with the CRISPR nuclease mRNA/gRNA mixture or with a preassembled RNPs (Ribonuclease protein of the CRISPR nuclease protein and gRNA), followed by cell washing. In embodiments of the present invention, the cells are suspended into a final formulation. In embodiments of the present invention, the cells may be resuspended. In embodiments of the present invention, the resuspended cells may be filled into bags for infusion. In embodiments of the present invention, the bags may be frozen using a freeze down step in a controlled rate freezer and/or stored in the vapor phase of liquid nitrogen. In embodiments of the present invention, the product may be administered by intravenous (IV) administration to a patient.

#### Dominant Genetic Disorders

[0306] One of skill in the art will appreciate that all subjects with any type of heterozygote genetic disorder (e.g., dominant genetic disorder) may be subjected to the methods described herein. In one embodiment, the present invention may be used to target a gene involved in, associated with, or causative of dominant genetic disorders such as, for example, SCN or CyN. In some embodiments, the dominant genetic disorder is SCN or CyN. In some embodiments, the target gene is the ELANE gene.

#### Crispr Nucleases and PAM Recognition

[0307] In some embodiments, the sequence specific nuclease is selected from CRISPR nucleases, or a functional variant thereof. In some embodiments, the sequence specific nuclease is an RNA guided DNA nuclease. In such embodiments, the RNA sequence which guides the RNA guided DNA nuclease (e.g., Cpf1) binds to and/or directs the RNA guided DNA nuclease to the sequence comprising at least one nucleotide which differs between a mutant allele and its counterpart functional allele (e.g., to a pathogenic mutation). In some embodiments, the CRISPR complex does not further comprise a tracrRNA. In a non-limiting example, in which the RNA guided DNA nuclease is a CRISPR protein, the at least one nucleotide which differs between the dominant mutant allele and the functional allele may be within the PAM site and/or proximal to the PAM site within the region that the RNA molecule is designed to hybridize to. A skilled artisan will appreciate that RNA molecules can be engineered to bind to a target of choice in a genome by commonly known methods in the art.

[0308] In embodiments of the present invention, a type II CRISPR system utilizes a mature crRNA:tracrRNA complex directs a CRISPR nuclease, e.g. Cas9, to the target DNA via Watson-Crick base-pairing between the crRNA and the protospacer on the target DNA next to the protospacer adjacent motif (PAM), an additional requirement for target recognition. The CRISPR nuclease then mediates cleavage of target DNA to create a double-stranded break within the protospacer. A skilled artisan will appreciate that each of the engineered RNA molecule of the present invention is further designed such as to associate with a target genomic DNA sequence of interest next to a protospacer adjacent motif (PAM), e.g., a PAM matching the sequence relevant for the type of CRISPR nuclease utilized, such as for a non-limiting example, NGG or NAG, wherein “N” is any nucleobase, for *Streptococcus pyogenes* Cas9 WT (SpCAS9); NNGRRT for *Staphylococcus aureus* (SaCas9); NNNVRYM for *Jejuni* Cas9 WT; NGAN or NGNG for SpCas9-

VQR variant; NGCG for SpCas9-VRER variant; NGAG for SpCas9-EQR variant; NNNNGATT for *Neisseria meningitidis* (NmCas9); or TTV for Cpf1. RNA molecules of the present invention are each designed to form complexes in conjunction with one or more different CRISPR nucleases and designed to target polynucleotide sequences of interest utilizing one or more different PAM sequences respective to the CRISPR nuclease utilized.

[0309] In some embodiments, an RNA-guided DNA nuclease e.g., a CRISPR nuclease, may be used to cause a DNA break at a desired location in the genome of a cell. The most commonly used RNA-guided DNA nucleases are derived from CRISPR systems, however, other RNA-guided DNA nucleases are also contemplated for use in the genome editing compositions and methods described herein. For instance, see U.S. Patent Publication No. 2015-0211023, incorporated herein by reference.

[0310] CRISPR systems that may be used in the practice of the invention vary greatly. CRISPR systems can be a type I, a type II, type III, or type V system. Non-limiting examples of suitable CRISPR proteins include Cas3, Cas4, Cas5, Cas5e (or CasD), Cas6, Cas6e, Cas6f, Cas7, Cas8a1, Cas8a2, Cas8b, Cas8c, Cas9, Cas10, Cas12a, Cas12b, Cas12c, Cas12d, Cas12d, Cas10d, CasF, CasG, CasH, Csy1, Csy2, Csy3, Cse1 (or CasA), Cse2 (or CasB), Cse3 (or CasE), Cse4 (or CasC), Csc1, Csc2, Csa5, Csn2, Csm2, Csm3, Csm4, Csm5, Csm6, Cmr1, Cmr3, Cmr4, Cmr5, Cmr6, Csb1, Csb2, Csb3, Csx17, Csx14, Csx10, Csx16, CsaX, Csx3, Csz1, Csx15, Csf1, Csf2, Csf3, Csf4, and Cu1966. (See. e.g., Koonin 2017).

[0311] In some embodiments, the RNA-guided DNA nuclease is a CRISPR nuclease derived from a type II CRISPR system (e.g., Cas9). The CRISPR nuclease may be derived from *Streptococcus pyogenes*, *Streptococcus thermophilus*, *Streptococcus* sp., *Staphylococcus aureus*, *Neisseria meningitidis*, *Treponema denticola*, *Nocardia* *dasdonvillei*, *Streptomyces pristinaespiralis*, *Streptomyces viridochromogenes*, *Streptomyces viridochromogenes*, *Streptosporangium roseum*, *Streptosporangium roseum*, *Alicyclobacillus acidocaldarius*, *Bacillus pseudomycoides*, *Bacillus selenitireducens*, *Exiguobacterium sibiricum*, *Lactobacillus delbrueckii*, *Lactobacillus salivarius*, *Microscilla marina*, *Burkholderiales bacterium*, *Polaromonas naphthalenivorans*, *Polaromonas* sp., *Crocospaera watsonii*, *Cyanothece* sp., *Microcystis aeruginosa*, *Synechococcus* sp., *Acetohalobium arabaticum*, *Ammonmfx degensii*, *Caldicelulosiruptor beccsii*, *Candidatus Desulfuridis*, *Clostridium botulinum*, *Clostridium difficile*, *Finnegoldia magna*, *Natranaerobius thermophilus*, *Pelotomaculumther mopropionicum*, *Acidithiobacillus caldus*, *Acidithiobacillus ferrooxidans*, *Allochrochromatium vinosum*, *Marinobacter* sp., *Nitrosococcus halophilus*, *Nitrosococcus watsoni*, *Pseudoalteromonas haloplanktis*, *Ktedonobacter racemifer*, *Methanohalobium evestigatum*, *Anabaena variabilis*, *Nodularia spumigena*, *Nostoc* sp., *Arthrospira maxima*, *Arthrospira platensis*, *Arthrospira* sp., *Lyngbya* sp., *Microcoleus chthonoplastes*, *Oscillatoria* sp., *Petrotoga mobilis*, *Thermosiphon africanus*, *Acaryochloris marina*, *Francisella* cf. *novicida* Fx1. *Alicyclobacillus acidoterrestris*, *Oleiphilus* sp., *Bacterium* CG09\_39\_24. *Deltaproteobacteria bacterium*, or any species which encodes a CRISPR nuclease with a known PAM sequence.

CRISPR nucleases encoded by uncultured bacteria may also be used in the context of the invention. (See Burstein et al. Nature, 2017). Variants of CRISPR proteins having known PAM sequences e.g., SpCas9 D1135E variant, SpCas9 VQR variant, SpCas9 EQR variant, or SpCas9 VRER variant may also be used in the context of the invention.

[0312] In embodiments of the present invention, the CRISPR nuclease has cleavage activity when used with an RNA molecule comprising a guide sequence portion having 21-23, or 21-22 nucleotides and may be derived from any of the species recited above. A specific example of a CRISPR nuclease which has cleavage activity when used with an RNA molecule comprising a guide sequence portion having 21 nucleotides is described in Mojica, F. J., et al. (1993). A specific example of a CRISPR nuclease which has cleavage activity when used with an RNA molecule comprising a guide sequence portion having 22 nucleotides is described in Kim, E. et al. (2017). Mojica, F. J., et al. (1993) and Kim, E. et al. (2017) in their entirety and/or for the specific

description CRISPR nucleases, are incorporated herein by reference. In such embodiments, the CRISPR nuclease may be engineered and/or non-naturally occurring.

[0313] In embodiments of the present invention, the CRISPR nuclease has greater cleavage activity when used with an RNA molecule comprising a guide sequence portion having 21-23 nucleotides, compared to its cleavage activity when used with an RNA molecule comprising a guide sequence portion having 20 or fewer nucleotides, and/or 24 or more nucleotides. In embodiments of the present invention, the CRISPR nuclease has greater cleavage activity when used with an RNA molecule comprising a guide sequence portion having 21-22 nucleotides, compared to its cleavage activity when used with an RNA molecule comprising a guide sequence portion having 20 or fewer nucleotides, and/or 23 or more nucleotides. In an embodiment, the CRISPR nuclease has its greatest cleavage activity when used with an RNA molecule comprising a guide sequence portion having 22 nucleotides.

[0314] Thus, an RNA guided DNA nuclease of a CRISPR system, such as a Cas9 protein or modified Cas9 or homolog or ortholog of Cas9, or other RNA guided DNA nucleases belonging to other types of CRISPR systems, such as Cpf1 and its homologs and orthologs, may be used in the compositions of the present invention.

[0315] In certain embodiments, the CRISPR nuclease may be a “functional derivative” of a naturally occurring Cas protein. A “functional derivative” of a native sequence polypeptide is a compound having a qualitative biological property in common with a native sequence polypeptide. “Functional derivatives” include, but are not limited to, fragments of a native sequence and derivatives of a native sequence polypeptide and its fragments, provided that they have a biological activity in common with a corresponding native sequence polypeptide. A biological activity contemplated herein is the ability of the functional derivative to hydrolyze a DNA substrate into fragments. The term “derivative” encompasses both amino acid sequence variants of polypeptide, covalent modifications, and fusions thereof. Suitable derivatives of a Cas polypeptide or a fragment thereof include but are not limited to mutants, fusions, covalent modifications of Cas protein or a fragment thereof. Cas protein, which includes Cas protein or a fragment thereof, as well as derivatives of Cas protein or a fragment thereof, may be obtainable from a cell or synthesized chemically or by a combination of these two procedures. The cell may be a cell that naturally produces Cas protein, or a cell that naturally produces Cas protein and is genetically engineered to produce the endogenous Cas protein at a higher expression level or to produce a Cas protein from an exogenously introduced nucleic acid, which nucleic acid encodes a Cas that is same or different from the endogenous Cas. In some cases, the cell does not naturally produce Cas protein and is genetically engineered to produce a Cas protein.

[0316] In some embodiments, the CRISPR nuclease is Cpf1. Cpf1 is a single RNA-guided endonuclease which utilizes a T-rich protospacer-adjacent motif. Cpf1 cleaves DNA via a staggered DNA double-stranded break. Two Cpf1 enzymes from *Acidaminococcus* and *Lachnospiraceae* have been shown to carry out efficient genome-editing activity in human cells. (See Zetsche et al. (2015) *Cell*.).

[0317] Thus, an RNA guided DNA nuclease of a Type II CRISPR System, such as a Cas9 protein or modified Cas9 or homologs, orthologues, or variants of Cas9, or other RNA guided DNA nucleases belonging to other types of CRISPR systems, such as Cpf1 and its homologs, orthologues, or variants, may be used in the present invention.

[0318] In some embodiments, the guide molecule comprises one or more chemical modifications which imparts a new or improved property (e.g., improved stability from degradation, improved hybridization energetics, or improved binding properties with an RNA guided DNA nuclease). Suitable chemical modifications include, but are not limited to one or more of: modified bases, modified sugar moieties, or modified inter-nucleoside linkages. Non-limiting examples of suitable chemical modifications include: 4-acetylcytidine, 5-(carboxyhydroxymethyl)uridine, 2'-O-methylcytidine, 5-carboxymethylaminomethyl-2-thiouridine, 5-carboxymethylaminomethyluridine,



dihydrouridine, 2'-O-methylpseudouridine, "beta, D-galactosylqueuosine", 2'-O-methylguanosine, inosine, N6-isopentenyladenosine, 1-methyladenosine, 1-methylpseudouridine, 1-methylguanosine, 1-methylinosine, "2,2-dimethylguanosine", 2-methyladenosine, 2-methylguanosine, 3-methylcytidine, 5-methylcytidine, N6-methyladenosine, 7-methylguanosine, 5-methylaminomethyluridine, 5-methoxyaminomethyl-2-thiouridine, "beta. D-mannosylqueosine", 5-methoxycarbonylmethyl-2-thiouridine, 5-methoxycarbonylmethyluridine, 5-methoxyuridine, 2-methylthio-N6-isopentenyladenosine, N-((9-beta-D-ribofuranosyl-2-methylthiopurine-6-yl)carbamoyl)threonine, N-((9-beta-D-ribofuranosylpurine-6-yl)N-methylcarbamoyl)threonine, uridine-5-oxyacetic acid-methylester, uridine-5-oxyacetic acid, wybutoxosine, queuosine, 2-thiocytidine, 5-methyl-2-thiouridine, 2-thiouridine, 4-thiouridine, 5-methyluridine, N-((9-beta-D-ribofuranosylpurine-6-yl)-carbamoyl)threonine, 2'-O-methyl-5-methyluridine, 2'-O-methyluridine, wybutosine, "3-(3-amino-3-carboxy-propyl)uridine, (acp3)u", 2'-O-methyl (M), 3'-phosphorothioate (MS), 3'-thioPACE (MSP), pseudouridine, or 1-methyl pseudo-uridine. Each possibility represents a separate embodiment of the present invention.

[0319] Further non-limiting examples of suitable chemical modifications include: m.sup.1A (1-methyladenosine); m.sup.2A (2-methyladenosine); Am (2'-O-methyladenosine); ms.sup.2m.sup.6A (2-methylthio-N.sup.6-methyladenosine); i.sup.6A (N.sup.6-isopentenyladenosine); ms.sup.2i6A (2-methylthio-N.sup.6isopentenyladenosine); io.sup.6A (N.sup.6-(cis-hydroxyisopentenyl)adenosine); ms.sup.2io.sup.6A (2-methylthio-N.sup.6-(cis-hydroxyisopentenyl) adenosine); g.sup.6A (N.sup.6-glycinyllcarbamoyl)adenosine); t.sup.6A (N.sup.6-threonyllcarbamoyl)adenosine); ms.sup.2t.sup.6A (2-methylthio-N.sup.6-threonyllcarbamoyl)adenosine); m.sup.6t.sup.6A (N.sup.6-methyl-N.sup.6-threonyllcarbamoyl)adenosine); hn.sup.6A (N.sup.6-hydroxynorvalyllcarbamoyl)adenosine); ms.sup.2hn.sup.6A (2-methylthio-N.sup.6-hydroxynorvalyllcarbamoyl)adenosine); Ar(p) (2'-O-ribosyladenosine (phosphate)); I (inosine); m.sup.1I (1-methylinosine); m.sup.1Im (1,2'-O-dimethylinosine); m.sup.3C (3-methylcytidine); Cm (2'-O-methylcytidine); s.sup.2C (2-thiocytidine); ac4C (N.sup.4-acetylcytidine); f.sup.5C (5-formylcytidine); m.sup.5Cm (5,2'-O-dimethylcytidine); ac.sup.4Cm (N.sup.4-acetyl-2'-O-methylcytidine); k.sup.2C (lysidine); m.sup.1G (1-methylguanosine); m.sup.2G (N.sup.2-methylguanosine); m.sup.7G (7-methylguanosine); Gm (2'-O-methylguanosine); m.sup.2.sub.2G (N.sup.2,N.sup.2-dimethylguanosine); m.sup.2Gm (N.sup.2,2'-O-dimethylguanosine); m.sup.22Gm (N.sup.2,N.sup.2, 2'-O-trimethylguanosine); Gr(p) (2'-O-ribosylguanosine (phosphate)); yW (wybutosine); .sub.o2yW (peroxywybutosine); OHyW (hydroxywybutosine); OHyW\* (undermodified hydroxywybutosine); imG (wyosine); mimG (methylwyosine); Q (queuosine); oQ (epoxyqueuosine); galQ (galactosyl-queuosine); manQ (mannosyl-queuosine); preQ.sub.0 (7-cyano-7-deazaguanosine); preQ.sub.1 (7-aminomethyl-7-deazaguanosine); G.sup.+ (archaeosine); D (dihydrouridine); m.sup.5Um (5,2'-O-dimethyluridine); s.sup.4U (4-thiouridine); m.sup.5s.sup.2U (5-methyl-2-thiouridine); s.sup.2Um (2-thio-2'-O-methyluridine); acp.sup.3U (3-(3-amino-3-carboxypropyl)uridine); ho.sup.5U (5-hydroxyuridine); mo.sup.5U (5-methoxyuridine); cmo.sup.5U (uridine 5-oxyacetic acid); mcmo.sup.5U (uridine 5-oxyacetic acid methyl ester); chm.sup.5U (5-(carboxyhydroxymethyl)uridine)); mchm.sup.5U (5-(carboxyhydroxymethyl)uridine methyl ester); mcm.sup.5U (5-methoxycarbonylmethyluridine); mcm.sup.5Um (5-methoxycarbonylmethyl-2'-O-methyluridine); mcm.sup.5s2U (5-methoxycarbonylmethyl-2-thiouridine); nm.sup.5S.sup.2U (5-aminomethyl-2-thiouridine); mnm.sup.5U (5-methylaminomethyluridine); mnm.sup.5s.sup.2U (5-methylaminomethyl-2-thiouridine); mnm.sup.5se.sup.2U (5-methylaminomethyl-2-selenouridine); ncm.sup.5U (5-carbamoylmethyluridine); ncm.sup.5Um (5-carbamoylmethyl-2'-O-methyluridine); cmnm.sup.5U (5-carboxymethylaminomethyluridine); cmnm.sup.5Um (5-carboxymethylaminomethyl-2'-O-methyluridine); cmmms.sup.2U (5-carboxymethylaminomethyl-2-thiouridine); dimethyladenosine); Im (2'-O-methylinosine); m.sup.4C (N4-methylcytidine); m.sup.4Cm (N.sup.4,2'-O-dimethylcytidine); hm.sup.5C (5-hydroxymethylcytidine); m.sup.3U (3-

methyluridine); cm.sup.5U (5-carboxymethyluridine); m.sup.6Am (N.sup.6,2'-O-dimethyladenosine); m.sup.6.sub.2Am (N.sup.6,N.sup.6,O-2'-trimethyladenosine); m.sup.2'7G (N.sup.2,7-dimethylguanosine); m2,2,7G (N.sup.2,N.sup.2,7-trimethylguanosine); m.sup.3Um (3,2'-O-dimethyluridine); m.sup.5D (5-methyldihydrouridine); f.sup.5Cm (5-formyl-2'-O-methylcytidine); m.sup.1 Gm (1,2'-O-dimethylguanosine); m.sup.1Am (1,2'-O-dimethyladenosine); tm.sup.5U (5-taurinomethyluridine); tm.sup.5s.sup.2U (5-taurinomethyl-2-thiouridine)); imG-14 (4-demethylwyosine); imG2 (isowyosine); or ac.sup.6A (N.sup.6-acetyladenosine). Each possibility represents a separate embodiment of the present invention. (See e.g. U.S. Pat. No. 9,750,824).

[0320] In embodiments of the present invention, the CRISPR nuclease has greater cleavage activity when used with an RNA molecule comprising a guide sequence portion having 21-23 nucleotides, compared to its cleavage activity when used with an RNA molecule comprising a guide sequence portion having 20 or fewer nucleotides, and/or 24 or more nucleotides. In embodiments of the present invention, the CRISPR nuclease has greater cleavage activity when used with an RNA molecule comprising a guide sequence portion having 21-22 nucleotides, compared to its cleavage activity when used with an RNA molecule comprising a guide sequence portion having 20 or fewer nucleotides, and/or 23 or more nucleotides. In an embodiment, the CRISPR nuclease has its greatest cleavage activity when used with an RNA molecule comprising a guide sequence portion having 22 nucleotides.

[0321] In an embodiment, such a CRISPR nuclease has at least 95% identity to the amino acid sequence as set forth in SEQ ID NO: 31125 or the sequence encoding the CRISPR nuclease has at least a 95% sequence identity to SEQ ID NO: 31131 or SEQ ID NO: 31132. In an embodiment, such a CRISPR nuclease has at least 100%, 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, 90%, 89%, 88%, 87%, 86%, 85%, 84%, 83%, or 82% identity to the amino acid sequence as set forth in SEQ ID NO: 31125 or the sequence encoding the CRISPR nuclease has at least a 100%, 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, 90%, 89%, 88%, 87%, 86%, 85%, 84%, 83%, or 82% sequence identity to SEQ ID NO: 31131 or SEQ ID NO: 31132.

#### Guide Sequences which Specifically Target a Mutant Allele

[0322] A given gene may contain thousands of sequence variations, including variations comprising a pathogenic mutation within a mutant allele. Utilizing a 24 base pair target window for specifically targeting each known pathogenic mutation in a mutant allele would require hundreds of thousands of guide sequences. Furthermore, any given guide sequence when utilized to target a specific sequence (e.g., a target sequence containing a pathogenic mutation) may result in degradation of the guide molecule, limited or no activity, or off-target effects. Accordingly, suitable guide sequences are necessary for targeting a given mutant allele. By the present invention, a novel set of guide sequences have been identified for knocking out expression of a mutated protein, inactivating a mutant ELANE gene allele, and treating SCN or CyN.

[0323] The present disclosure provides guide sequences capable of specifically targeting a mutant allele for inactivation while leaving the functional allele unmodified. The guide sequences of the present invention are designed to, and are most likely to, specifically differentiate between a mutant allele and a functional allele. Of all possible guide sequences which target a mutant allele desired to be inactivated, the specific guide sequences disclosed herein are specifically effective to function with the disclosed embodiments.

[0324] For each gene, any one of the following strategies may be used to deactivate the mutant allele: (1) Knockout strategy using one RNA molecule—one RNA molecule is utilized to direct a CRISPR nuclease to a mutant allele and create a double-strand break (DSB) leading to formation of a frameshift mutation in an exon or in a splice site region of the mutant allele; (2) Knockout strategy using two RNA molecules—two RNA molecules are utilized. A first RNA molecule targets a region in the promoter or an upstream region of a mutant allele and another RNA molecule targets downstream of the first RNA molecule in a promoter, exon, or intron of the mutant allele;

(3) Exon(s) skipping strategy—one RNA molecule may be used to target a CRISPR nuclease to a splice site region, either at the 5' end of an intron (donor sequence) or the 3' end of an intron (acceptor sequence), in order to destroy the splice site. Alternatively, two RNA molecules may be utilized such that a first RNA molecule targets an upstream region of an exon and a second RNA molecule targets a region downstream of the first RNA molecule, thereby excising the exon(s). Based on the locations of each mutant allele, any one of, or a combination of, the above-mentioned methods to deactivate the mutant allele may be utilized.

[0325] The at least one nucleotide which differs between the mutant allele and the functional allele, may be upstream, downstream or within the sequence of the disease-causing mutation of the gene of interest. The at least one nucleotide which differs between the mutant allele and the functional allele, may be within an exon or within an intron of the gene of interest. In some embodiments, the at least one nucleotide which differs between the mutant allele and the functional allele is within an exon of the gene of interest. In some embodiments, the at least one nucleotide which differs between the mutant allele and the functional allele is within an intron or an exon of the gene of interest, in close proximity to a splice site between the intron and the exon e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 nucleotides upstream or downstream to the splice site.

[0326] Guide sequences of the present invention may satisfy any one of the above criteria and are most likely to differentiate between a mutant allele from its corresponding functional allele.

[0327] In some embodiments the RNA molecule targets a mutant allele in the ELANE gene as shown in column 1 of Table 1 below.

[0328] Embodiments of the present invention may include excising the promoter region until intron 3 or intron 4 or the 3' UTR.

[0329] Embodiments of the present invention may include excising from intron 3 or intron 4 or 3' UTR to regions downstream to the 3' UTR.

[0330] Embodiments of the present invention excising from intron 3 or intron 4 or 3' UTR to regions downstream to the 3' UTR. The strategy is designed such as to specifically knock-out the disease-causing allele ('mutant allele'), while leaving the healthy allele intact.

#### Delivery to Cells

[0331] It is understood that in the methods embodied, the RNA molecules and compositions described herein may be delivered to a target cell or subject by any suitable means. The following embodiments provide non-limiting examples of methods of delivery of the RNA molecules and composition of the present invention.

[0332] In some embodiments, RNA molecule compositions of the present invention may be targeted to any cell which contains and/or expresses a dominant negative allele, including any mammalian or plant cell. For example, in one embodiment the RNA molecule specifically targets a mutant ELANE allele and the target cell is a hepatocyte cell.

[0333] Any suitable viral vector system may be used to deliver nucleic acid compositions e.g., the RNA molecule compositions of the subject invention. Conventional viral and non-viral based gene transfer methods can be used to introduce nucleic acids and target tissues. In certain embodiments, nucleic acids are administered for in vivo or ex vivo gene therapy uses. Non-viral vector delivery systems include naked nucleic acid, and nucleic acid complexed with a delivery vehicle such as a liposome or poloxamer. For a review of gene therapy procedures, see Anderson (1992) Science 256:808-813; Nabel & Felgner (1993) TIBTECH 11:211-217; Mitani & Caskey (1993) TIBTECH 11:162-166; Dillon (1993) TIBTECH 11:167-175; Miller (1992) Nature 357:455-460; Van Brunt (1988) Biotechnology 6(10):1149-1154; Vigne (1995) Restorative Neurology and Neuroscience 8:35-36; Kremer & Perricaudet (1995) British Medical Bulletin 51(1):31-44; Haddada et al. (1995) in Current Topics in Microbiology and Immunology Doerfler and Bohm (eds.); and Yu et al. (1994) Gene Therapy 1:13-26.

[0334] Methods of non-viral delivery of nucleic acids and/or proteins include electroporation,

lipofection, microinjection, biolistics, particle gun acceleration, virosomes, liposomes, immunoliposomes, lipid nanoparticles (LNPs), polycation or lipid:nucleic acid conjugates, artificial virions, and agent-enhanced uptake of nucleic acids or can be delivered to plant cells by bacteria or viruses (e.g., *Agrobacterium*, *Rhizobium* sp. NGR234, *Sinorhizobium meliloti*, *Mesorhizobium loti*, tobacco mosaic virus, potato virus X, cauliflower mosaic virus and cassava vein mosaic virus). (See, e.g., Chung et al. (2006) Trends Plant Sci. 11(1):1-4). Sonoporation using, e.g., the Sonitron 2000 system (Rich-Mar), can also be used for delivery of nucleic acids. Cationic-lipid mediated delivery of proteins and/or nucleic acids is also contemplated as an in vivo or in vitro delivery method. (See Zuris et al. (2015) Nat. Biotechnol. 33(1):73-80; see also Coelho et al. (2013) N. Engl. J. Med. 369, 819-829; Judge et al. (2006) Mol. Ther. 13, 494-505; and Basha et al. (2011) Mol. Ther. 19, 2186-2200).

[0335] Additional exemplary nucleic acid delivery systems include those provided by Amaxa® Biosystems (Cologne, Germany), Maxcyte, Inc. (Rockville, Md.), BTX Molecular Delivery Systems (Holliston, Mass.) and Copernicus Therapeutics Inc., (see, e.g., U.S. Pat. No. 6,008,336). Lipofection is described in e.g., U.S. Pat. Nos. 5,049,386, 4,946,787; and 4,897,355, and lipofection reagents are sold commercially (e.g., Transfectam™, Lipofectin™ and Lipofectamine™ RNAiMAX). Cationic and neutral lipids that are suitable for efficient receptor-recognition lipofection of polynucleotides include those of Felgner, WO 91/17424, WO 91/16024. Delivery can be to cells (ex vivo administration) or target tissues (in vivo administration).

[0336] The preparation of lipid:nucleic acid complexes, including targeted liposomes such as immunolipid complexes, is well known to one of skill in the art (See, e.g., Crystal (1995) Science 270:404-410; Blaese et al. (1995) Cancer Gene Ther. 2:291-297; Behr et al. (1994) Bioconjugate Chem. 5:382-389; Remy et al. (1994) Bioconjugate Chem. 5:647-654; Gao et al. (1995) Gene Therapy 2:710-722; Ahmad et al. (1992) Cancer Res. 52:4817-4820; U.S. Pat. Nos. 4,186,183, 4,217,344, 4,235,871, 4,261,975, 4,485,054, 4,501,728, 4,774,085, 4,837,028, and 4,946,787).

[0337] Additional methods of delivery include the use of packaging the nucleic acids to be delivered into EnGeneIC delivery vehicles (EDVs). These EDVs are specifically delivered to target tissues using bispecific antibodies where one arm of the antibody has specificity for the target tissue and the other has specificity for the EDV. The antibody brings the EDVs to the target cell surface and then the EDV is brought into the cell by endocytosis. Once in the cell, the contents are released (See MacDiarmid et al (2009) Nature Biotechnology 27(7):643).

[0338] The use of RNA or DNA viral based systems for viral mediated delivery of nucleic acids take advantage of highly evolved processes for targeting a virus to specific cells in the body and trafficking the viral payload to the nucleus. Viral vectors can be administered directly to patients (in vivo) or they can be used to treat cells in vitro and the modified cells are administered to patients (ex vivo). Conventional viral based systems for the delivery of nucleic acids include, but are not limited to, retroviral, lentivirus, adenoviral, adeno-associated, vaccinia and herpes simplex virus vectors for gene transfer.

[0339] The tropism of a retrovirus can be altered by incorporating foreign envelope proteins, expanding the potential target population of target cells. Lentiviral vectors are retroviral vectors that are able to transduce or infect non-dividing cells and typically produce high viral titers. Selection of a retroviral gene transfer system depends on the target tissue. Retroviral vectors are comprised of cis-acting long terminal repeats with packaging capacity for up to 6-10 kb of foreign sequence. The minimum cis-acting LTRs are sufficient for replication and packaging of the vectors, which are then used to integrate the therapeutic gene into the target cell to provide permanent transgene expression. Widely used retroviral vectors include those based upon murine leukemia virus (MuLV), gibbon ape leukemia virus (GaLV), Simian Immunodeficiency virus (SIV), human immunodeficiency virus (HIV), and combinations thereof (See, e.g., Buchschacher et al. (1992) J. Virol. 66:2731-2739; Johann et al. (1992) J. Virol. 66:1635-1640; Sommerfelt et al. (1990) Virol. 176:58-59; Wilson et al. (1989) J. Virol. 63:2374-2378; Miller et al. (1991) J. Virol. 65:2220-2224;

[0340] At least six viral vector approaches are currently available for gene transfer in clinical trials, which utilize approaches that involve complementation of defective vectors by genes inserted into helper cell lines to generate the transducing agent.

[0341] pLASN and MFG-S are examples of retroviral vectors that have been used in clinical trials (Dunbar et al. (1995) *Blood* 85:3048-305; Kohn et al. (1995) *Nat. Med.* 1:1017-102; Malech et al. (1997) *PNAS* 94:22 12133-12138). PA317/pLASN was the first therapeutic vector used in a gene therapy trial. (Blaese et al. (1995). Transduction efficiencies of 50% or greater have been observed for MFG-S packaged vectors. (Ellem et al. (1997) *Immunol Immunother*, 44(1):10-20; Dranoff et al. (1997) *Hum. Gene Ther.* 1:111-2).

[0342] Packaging cells are used to form virus particles that are capable of infecting a host cell. Such cells include 293 cells, which package adenovirus, AAV, and Psi-2 cells or PA317 cells, which package retrovirus. Viral vectors used in gene therapy are usually generated by a producer cell line that packages a nucleic acid vector into a viral particle. The vectors typically contain the minimal viral sequences required for packaging and subsequent integration into a host (if applicable), other viral sequences being replaced by an expression cassette encoding the protein to be expressed. The missing viral functions are supplied in trans by the packaging cell line. For example, AAV vectors used in gene therapy typically only possess inverted terminal repeat (ITR) sequences from the AAV genome which are required for packaging and integration into the host genome. Viral DNA is packaged in a cell line, which contains a helper plasmid encoding the other AAV genes, namely rep and cap, but lacking ITR sequences. The cell line is also infected with adenovirus as a helper. The helper virus promotes replication of the AAV vector and expression of AAV genes from the helper plasmid. The helper plasmid is not packaged in significant amounts due to a lack of ITR sequences. Contamination with adenovirus can be reduced by, e.g., heat treatment to which adenovirus is more sensitive than AAV. Additionally, AAV can be produced at clinical scale using baculovirus systems (see U.S. Pat. No. 7,479,554).

[0343] In many gene therapy applications, it is desirable that the gene therapy vector be delivered with a high degree of specificity to a particular tissue type. Accordingly, a viral vector can be modified to have specificity for a given cell type by expressing a ligand as a fusion protein with a viral coat protein on the outer surface of the virus. The ligand is chosen to have affinity for a receptor known to be present on the cell type of interest. For example, Han et al. (1995) *Proc. Natl. Acad. Sci. USA* 92:9747-9751, reported that Moloney murine leukemia virus can be modified to express human heregulin fused to gp70, and the recombinant virus infects certain human breast cancer cells expressing human epidermal growth factor receptor. This principle can be extended to other virus-target cell pairs, in which the target cell expresses a receptor and the virus expresses a fusion protein comprising a ligand for the cell-surface receptor. For example, filamentous phage can be engineered to display antibody fragments (e.g., FAB or Fv) having specific binding affinity for virtually any chosen cellular receptor. Although the above description applies primarily to viral vectors, the same principles can be applied to nonviral vectors. Such vectors can be engineered to contain specific uptake sequences which favor uptake by specific target cells.

[0344] Gene therapy vectors can be delivered in vivo by administration to an individual patient, typically by systemic administration (e.g., intravitreal, intravenous, intraperitoneal, intramuscular, subdermal, or intracranial infusion) or topical application, as described below. Alternatively, vectors can be delivered to cells ex vivo, such as cells explanted from an individual patient (e.g., lymphocytes, bone marrow aspirates, tissue biopsy) or universal donor hematopoietic stem cells, followed by reimplantation of the cells into a patient, usually after selection for cells which have incorporated the vector.

[0345] Ex vivo cell transfection for diagnostics, research, or for gene therapy (e.g., via re-infusion of the transfected cells into the host organism) is well known to those of skill in the art. In a preferred embodiment, cells are isolated from the subject organism, transfected with a nucleic acid

composition, and re-infused back into the subject organism (e.g., patient). Various cell types suitable for ex vivo transfection are well known to those of skill in the art (See, e.g., Freshney et al. (1994) Culture of Animal Cells, A Manual of Basic Technique, 3rd ed, and the references cited therein for a discussion of how to isolate and culture cells from patients).

[0346] Suitable cells include, but are not limited to, eukaryotic cells and/or cell lines. Non-limiting examples of such cells or cell lines generated from such cells include COS, CHO (e.g., CHO-S, CHO-K1, CHO-DG44, CHO-DUXB11, CHO-DUKX, CHOK1SV), VERO, MDCK, WI38, V79, B14AF28-G3, BHK, HaK, NSO, SP2/0-Ag14, HeLa, HEK293 (e.g., HEK293-F, HEK293-H, HEK293-T), perC6 cells, any plant cell (differentiated or undifferentiated), as well as insect cells such as *Spodoptera fugiperda* (Sf), or fungal cells such as *Saccharomyces*, *Pichia* and *Schizosaccharomyces*. In certain embodiments, the cell line is a CHO-K1, MDCK or HEK293 cell line. Additionally, primary cells may be isolated and used ex vivo for reintroduction into the subject to be treated following treatment with a guided nuclease system (e.g. CRISPR/Cas). Suitable primary cells include peripheral blood mononuclear cells (PBMC), and other blood cell subsets such as, but not limited to, CD4<sup>+</sup> T cells or CD8<sup>+</sup> T cells. Suitable cells also include stem cells such as, by way of example, embryonic stem cells, induced pluripotent stem cells, hematopoietic stem cells (CD34<sup>+</sup>), neuronal stem cells and mesenchymal stem cells.

[0347] In one embodiment, stem cells are used in ex vivo procedures for cell transfection and gene therapy. The advantage to using stem cells is that they can be differentiated into other cell types in vitro, or can be introduced into a mammal (such as the donor of the cells) where they will engraft in the bone marrow. Methods for differentiating CD34<sup>+</sup> cells in vitro into clinically important immune cell types using cytokines such as GM-CSF, IFN-gamma, and TNF-alpha are known (as a non-limiting example see, Inaba et al., J. Exp. Med. 176:1693-1702 (1992)).

[0348] Stem cells are isolated for transduction and differentiation using known methods. For example, stem cells are isolated from bone marrow cells by panning the bone marrow cells with antibodies which bind unwanted cells, such as CD4<sup>+</sup> and CD8<sup>+</sup> (T cells), CD45<sup>+</sup> (panB cells), GR-1 (granulocytes), and Iad (differentiated antigen presenting cells) (as a non-limiting example see Inaba et al. (1992) J. Exp. Med. 176:1693-1702). Stem cells that have been modified may also be used in some embodiments.

[0349] Typically, the cells are administered in a pharmaceutical composition comprising at least one pharmaceutically-acceptable carrier. The phrase “pharmaceutically acceptable” refers to those compounds, materials, compositions, and/or dosage forms which are, within the scope of sound medical judgment, suitable for use in contact with the tissues of human beings and animals without excessive toxicity, irritation, allergic response, or other problem or complication, commensurate with a reasonable benefit/risk ratio. The phrase “pharmaceutically acceptable carrier” as used herein means a pharmaceutically-acceptable material, composition or vehicle, such as a liquid or solid filler, diluent, excipient, or solvent encapsulating material.

[0350] Any one of the RNA molecule compositions described herein is suitable for genome editing in post-mitotic cells or any cell which is not actively dividing, e.g., arrested cells. Examples of post-mitotic cells which may be edited using an RNA molecule composition of the present invention include, but are not limited to, a hepatocyte cell.

[0351] Vectors (e.g., retroviruses, liposomes, etc.) containing therapeutic nucleic acid compositions can also be administered directly to an organism for transduction of cells in vivo. Administration is by any of the routes normally used for introducing a molecule into ultimate contact with blood or tissue cells including, but not limited to, injection, infusion, topical application (e.g., eye drops and cream) and electroporation. Suitable methods of administering such nucleic acids are available and well known to those of skill in the art, and, although more than one route can be used to administer a particular composition, a particular route can often provide a more immediate and more effective reaction than another route. According to some embodiments, the composition is delivered via IV injection.

[0352] Vectors suitable for introduction of transgenes into immune cells (e.g., T-cells) include non-integrating lentivirus vectors. See, e.g., U.S. Patent Publication No. 2009-0117617.

[0353] Pharmaceutically acceptable carriers are determined in part by the particular composition being administered, as well as by the particular method used to administer the composition. Accordingly, there is a wide variety of suitable formulations of pharmaceutical compositions available, as described below (See, e.g., Remington's Pharmaceutical Sciences, 17th ed., 1989).

[0354] In accordance with some embodiments, there is provided an RNA molecule which binds to/associates with and/or directs the RNA guided DNA nuclease to a sequence comprising at least one nucleotide which differs between a mutant allele and a functional allele (e.g., a pathogenic mutation) of a gene of interest. For example, a disease-causing or pathogenic sequence of the mutant allele may be specifically targeted since this sequence is not present in the functional allele.

[0355] The disclosed compositions and methods may also be used in the manufacture of a medicament for treating dominant genetic disorders in a patient.

#### Mechanisms of Action for Several Embodiments Disclosed Herein

[0356] Mutations in ELANE that were demonstrated to lead to SCN or CyN, mediate translation from alternative in frame ORF (open reading frame) that generate truncated N-terminus protein thus causing ER and protein misfolding stress.

[0357] Without being bound by any theory or mechanism, the instant invention may be utilized to apply a CRISPR nuclease to process the mutated pathologic ELANE allele and not the functional ELANE allele, such as to prevent expression of the mutated pathologic allele or to produce a truncated non-pathologic peptide from the mutated pathologic allele, or to repair/correct the mutated pathologic ELANE allele in order to prevent ameliorate or treat SCN or CyN.

[0358] Several alternative editing strategies include exclusion of the whole gene, truncation of the gene to exclude the C-terminus of the gene, and attenuation of the expression of the gene.

#### Examples of RNA Guide Sequences which Specifically Target Mutant Alleles of ELANE Gene.

[0359] Although a large number of guide sequences can be designed to target a mutant allele, the nucleotide sequences described in Table 1 identified by SEQ ID NOs: 1-31117 below were specifically selected to effectively implement the methods set forth herein and to effectively discriminate between alleles.

[0360] Table 1 shows guide sequences designed for use as described in the embodiments above to associate with mutant ELANE alleles. Each engineered guide molecule is further designed such as to associate with a target genomic DNA sequence of interest that lies next to a protospacer adjacent motif (PAM), e.g., a PAM matching the sequence NGG or NAG, where "N" is any nucleobase. The guide sequences were designed to work in conjunction with one or more different CRISPR nucleases, including, but not limited to, e.g. SpCas9WT (PAM SEQ: NGG), SpCas9.VQR1 (PAM SEQ: NGAN), SpCas9.VQR2 (PAM SEQ: NGNG), SpCas9.EQR (PAM SEQ: NGAG), SpCas9.VRER (PAM SEQ: NGCG), SaCas9WT (PAM SEQ: NNGRRT), NmCas9WT (PAM SEQ: NNNNGATT), Cpf1 (PAM SEQ: TITV), or JeCas9WT (PAM SEQ: NNNVRYM). RNA molecules of the present invention are each designed to form complexes in conjunction with one or more different CRISPR nucleases and designed to target polynucleotide sequences of interest utilizing one or more different PAM sequences respective to the CRISPR nuclease utilized.

TABLE-US-00002 TABLE 1 Guide sequences designed to target mutant alleles the ELANE gene

| SEQ ID NOs:      | base guides       | SEQ ID NOs:                                                                               | base guides    | SEQ ID NOs: | base guides | Target               |                                                      |
|------------------|-------------------|-------------------------------------------------------------------------------------------|----------------|-------------|-------------|----------------------|------------------------------------------------------|
| 19:              | 852329_A_G        | 1-41                                                                                      | 42-63          | 64-95       | 19:         |                      |                                                      |
| 852330_T_G       | 1, 20, 31, 96-135 | 136-157                                                                                   | 65-66, 158-188 | 19:         | 852331_G_C  | 31, 97, 114, 229-269 |                                                      |
| 65, 158, 270-312 | 189-228           | 19:                                                                                       | 852338_G_A     | 313-358     | 359-406     | 407-456              |                                                      |
| 19:              | 852342_G_A        | 457-502                                                                                   | 503-550        | 551-600     | 19:         | 852343_CCG_CCGCC     | 313-315, 344, 359-360, 391, 407, 417, 440, G 601-620 |
| 400, 621-632     | 450, 633-652      | 19:                                                                                       | 852350_G_A     | 653-698     | 699-746     | 747-796              |                                                      |
| 19:              | 852351_C_T        | 656, 665-666, 702, 712, 753, 759, 670, 684, 698, 717-718, 731, 767-768, 781, 790, 797-836 | 740, 837-878   | 879-922     | 19:         | 852366_C_G           | 923-                                                 |

968 969-1016 1017-1066 19: 852367\_C\_G 926-927, 933, 974, 980, 983, 1022, 1029, 939, 948, 950, 986, 995, 997, 1032, 1035, 1067-1106 1107-1148 1047, 1049, 1149-1192 19: 852380\_G\_C 1193-1238 1239-1284 1285-1334 19: 852392\_G\_C 1335-1380 1381-1426 1427-1476 19: 852395\_GG\_G 1369, 1377, 1411, 1414, 1434, 1461, 1380, 1477-1509 1426, 1510-1544 1464, 1545-1581 19: 852396\_G\_T 1486-1487, 1492, 1520, 1525, 1555, 1560, 1497, 1505, 1530, 1538-1539, 1571, 1574-1575, 1582-1622 1623-1665 1666-1710 19: 852864\_G\_T 1711-1756 1757-1803 1804-1852 19: 852878\_A\_C 1853-1880 1881-1896 1897-1918 19: 852882\_C\_T 1919-1964 1965-2012 2013-2062 19: 852885\_T\_C 1940, 1946, 1988, 1991, 2037, 2050, 2063-2088 2089-2112 2113-2140 19: 852898\_T\_G 2141-2179 2180-2211 2212-2248 19: 852927\_C\_T 2249-2283 2284-2310 2311-2339 19: 852933\_C\_T 2340-2378 2379-2417 2418-2458 19: 852935\_T\_C 2350, 2364, 2395, 2403, 2434, 2446, 2459-2493 2494-2528 2529-2565 19: 852937\_C\_A 2476, 2484, 2512, 2520, 2546, 2548, 2566-2606 2607-2648 2649-2692 19: 852939\_T\_G 2582, 2584, 2623, 2644, 2670, 2688, 2586, 2693-2734 2735-2777 2778-2822 19: 852941\_GTGTCCC\_G 2823-2863 2864-2906 2907-2951 19: 852941\_G\_A 2714, 2720, 2751, 2758, 2795, 2801, 2726, 2731, 2770, 2875, 2810, 2934, 2832, 2952-2992 2993-3034 3035-3079 19: 852941\_G\_C 2714, 2720, 2751, 2758, 2795, 2801, 2726, 2731, 2770, 2875, 2810, 2934, 2832, 2964, 3019, 3021, 3068, 3070, 3118-3157 3078, 3158-3199 2967, 2980, 3080-3117 19: 852941\_G\_T 2714, 2720, 2751, 2758, 2795, 2801, 2726, 2731, 2770, 2875, 2810, 2934, 2832, 2964, 3019, 3021, 3068, 3070, 2967, 2980, 3238-3277 3078, 3278-3319 3200-3237 19: 852942\_T\_A 2714, 2720, 2751, 2770, 2801, 2810, 2831-2832, 2964, 2874-2875, 3019, 2919, 2934, 2967, 2980, 3021, 3359-3399 3068, 3070, 3320-3358 3078, 3400-3442 19: 852945\_C\_A 2824, 2826, 2864, 2868, 2907, 2917-2919, 2828, 2831, 2870, 2873-2874, 2931, 3436, 2856, 3350, 3398, 3483-3524 3525-3568 3443-3482 19: 852945\_C\_G 2824, 2826, 2864, 2868, 2907, 2917-2919, 2828, 2831, 2870, 2873-2874, 2931, 3436, 2856, 3350, 3398, 3483, 3528, 3530, 3455, 3479, 3515, 3607-3645 3646-3687 3569-3606 19: 852945\_C\_T 2824, 2826, 2864, 2868, 2907, 2917-2919, 2828, 2831, 2870, 2873-2874, 2931, 3436, 2856, 3350, 3398, 3483, 3528, 3530, 3455, 3479, 3515, 3726-3765 3766-3807 3688-3725 19: 852948\_T\_C 2823, 2826, 2864, 2866, 2909, 2914, 2844, 3455, 2887, 3483, 2931, 3530, 3808-3845 3846-3877 3878-3915 19: 852948\_T\_G 2823, 2826, 2864, 2866, 2909, 2914, 2844, 3455, 2887, 3483, 2931, 3530, 3808, 3821, 3849, 3856, 3879, 3892, 3829, 3834, 3858, 3866, 3898, 3903, 3916-3949 3950-3977 3978-4011 19: 852954\_T\_C 4012-4053 4054-4089 4090-4131 19: 852965\_C\_T 4132-4177 4178-4225 4226-4275 19: 852966\_A\_T 4133, 4139, 4179, 4185-4186, 4227, 4234, 4143, 4169, 4216, 4221, 4250, 4266, 4172-4173, 4276- 4225, 4316-4356 4268, 4275, 4315 4357-4400 19: 852967\_C\_A 4133, 4139, 4179, 4186, 4234, 4250, 4143, 4173, 4216, 4221, 4266, 4275, 4282, 4288, 4338, 4441-4482 4371, 4397, 4401-4440 4483-4526 19: 852971\_T\_A 4527-4572 4573-4620 4621-4670 19: 852972\_G\_A 4529, 4536, 4582, 4597, 4641, 4646, 4550, 4553, 4608, 4610, 4659, 4665, 4561, 4563, 4616, 4619, 4668, 4670, 4671-4710 4711-4752 4753-4796 19: 852972\_G\_C 4529, 4536, 4582, 4597, 4641, 4646, 4550, 4553, 4608, 4610, 4659, 4665, 4561, 4563, 4616, 4619, 4668, 4670, 4696, 4709, 4713, 4752, 4761, 4790, 4797-4834 4835-4874 4875-4916 19: 852974\_G\_C 4529, 4563, 4582, 4619, 4646, 4670, 4696, 4709, 4713, 4752, 4761, 4790, 4917-4958 4959-5002 5003-5048 19: 852976\_C\_CCTGATT 4955, 5049-5092 4976, 5093-5138 5033, 5139-5186 19: 852977\_G\_A 4951, 4955, 4963, 4976, 5003, 5033, 5064, 5075-5076, 5108, 5119, 5157, 5166, 5187-5227 5135, 5228-5270 5182, 5271-5315 19: 852977\_G\_T 4951, 4955, 4963, 4976, 5003, 5033, 5064, 5075-5076, 5108, 5119, 5157, 5166, 5187, 5192, 5135, 5228, 5182, 5272, 5204, 5316-5353 5239, 5245, 5283, 5290, 5354-5393 5394-5435 19: 852978\_C\_T 5064, 5075-5076, 5108, 5119, 5157, 5166, 5187, 5192, 5135, 5228, 5182, 5272, 5204, 5436-5475 5239, 5245, 5283, 5290, 5476-5517 5518-5561 19: 852980\_A\_C 5076, 5204, 5108,



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30938 19: 856144\_C\_T 30939-30974 30975-31001 31002-31030 19: 856145\_C\_T 30953,  
30970, 30986, 30998, 31009, 31015, 30974, 31031-31063 31001, 31030, 31090-31117  
31064-31089 The indicated locations listed in column 1 of the Table 1 are based on gnomAD v3  
database and UCSC Genome Browser assembly ID: hg38, Sequencing/Assembly provider ID:  
Genome Reference Consortium Human GRCh38.p12 (GCA\_000001405.27). Assembly date:  
December 2013 initial release; December 2017 patch release 12. The SNP details are indicated by  
the listed SNP ID NOs. ("rs numbers"), which are based on the NCBI 2018 database of Single  
Nucleotide Polymorphisms (dbSNP)). The indicated DNA mutations are associated with Transcript  
Consequence NM\_001972 as obtained from NCBI RefSeq genes.

[0361] For the foregoing embodiments, each embodiment disclosed herein is contemplated as being applicable to each of the other disclosed embodiment. For example, it is understood that any of the RNA molecules or compositions of the present invention may be utilized in any of the methods of the present invention.

[0362] As used herein, all headings are simply for organization and are not intended to limit the disclosure in any manner. The content of any individual section may be equally applicable to all sections.

[0363] Examples are provided below to facilitate a more complete understanding of the invention. The following examples illustrate the exemplary modes of making and practicing the invention. However, the scope of the invention is not limited to specific embodiments disclosed in these Examples, which are for purposes of illustration only.

[0364] Further, the examples herein below disclose methods utilizing SpCas9 and guide sequences suitable to target the SpCas9 to the disclosed ELANE pathogenic mutation sequences. The examples demonstrate the feasibility of the strategies disclosed. A person having ordinary skill in the art would understand that the same guides sequences may be used with different CRISPR

nuclease to target the disclosed mutations to apply each of the specified strategies. Further, different guide sequences that target other CRISPR nucleases to the same mutations may be used together with the other nucleases to apply each of the specified strategies.

## EXPERIMENTAL DETAILS

[0365] Examples are provided below to facilitate a more complete understanding of the invention. The following examples illustrate the exemplary modes of making and practicing the invention. However, the scope of the invention is not limited to specific embodiments disclosed in these Examples, which are for purposes of illustration only.

[0366] CRISPR repeat (crRNA), transactivating crRNA (tracrRNA), nuclease polypeptide sequences, and PAM sequences were predicted from different metagenomic databases of sequences of environmental samples. The list of bacterial species/strains from which the CRISPR repeat, tracrRNA and nuclease sequences were predicted is provided in Table 2.

TABLE-US-00003 TABLE 2 DNA and Protein Sequences of Nucleases Human Original optimized Protein ORF DNA DNA Sequence Sequence Sequence Name Source Organism SEQ ID NO: SEQ ID NO: SEQ ID NO: OMNI-39 *Butyrivibrio* sp. 31123 31127 31128 AC2005 OMNI-40 *Aliiarcobacter* 31124 31129 31130 *faecis* OMNI-50 *Ezakiella* 31125 31131 31132 *peruensis* strain M6.X2 OMNI-53 *Clostridium* sp. 31126 31133 31134 AF02-29

### Construction of OMNI Nuclease Polypeptides

[0367] For construction of OMNI nuclease polypeptides, the open reading frame of each nuclease (OMNI) that was identified was codon optimized for *E. coli* and for human cell line expression. The *E. coli* optimized ORF was cloned into bacterial plasmid pb-NNC and the human optimized into the pmOMNI plasmid (Table 3).

TABLE-US-00004 TABLE 3 Plasmids and Constructs SEQ ID NO Plasmid Purpose Elements Example (DNA Sequence) pbNNC-2 Expressing OMNI T7 promoter HA Tag-Linker-OMNI pbNNC2 OMNI39 31135 polypeptide in the ORF (Human optimized) - T7 bacterial system terminator pbGuide Expressing OMNI J23119 promoter - T1/T2 spacer pbGuide OMNI39 31136 T1/T2 sgRNA in the sgRNA scaffold - rrnB T1 terminator T2 sgRNA V2 bacterial system pbPOS T2 Bacterial/TXTL T2 protospacer - 8N PAM library - pbPOS T2 library 31137 library depletion assay chloramphenicol acetyltransferase pET9a Expression and T7 promoter - SV40 NLS - OMNI pET9a OMNI50- 31138 purification of ORF (human optimized) - HA - SV40 HisTag OMNI proteins NLS - 8 His-tag - T7 terminator pmOMNI Expressing OMNI CMV promoter - Kozak - SV40 NLS - pmOMNI OMNI39 31139 polypeptide in the OMNI ORF (human optimized) - HA - mammalian system SV40 NLS - P2A - mCherry - bGH poly(A) signal pmGuide Expressing OMNI U6 promoter - Endogenic spacer pmGuide OMNI39 31140 Endogenic sgRNA in the sgRNA scaffold CXCR4 sgRNA V3 site mammalian system

TABLE-US-00005 TABLE 3 APPENDIX Details of construct elements Element Protein Sequence DNA sequence HA Tag SEQ ID NO: 31141 SEQ ID NO: 31142 NLS SEQ ID NO: 31143 SEQ ID NO: 31144 P2A SEQ ID NO: 31145 SEQ ID NO: 31146 mCherry SEQ ID NO: 31147 SEQ ID NO: 31148

### Prediction and Construction of sgRNA.

[0368] For each OMNI the sgRNA was predicted by detection of the CRISPR repeat array sequence (crRNA) and a trans-activating crRNA (tracrRNA) in the bacterial genome. The native pre-mature crRNA and tracrRNA sequences were connected in-silico with tetra-loop 'gaaa' and the secondary structure elements of the duplex were predicted by using an RNA Secondary Structure prediction Web Tool available at [ma.urmc.rochester.edu](http://ma.urmc.rochester.edu).

[0369] The predicted secondary structures of the full duplex RNA elements (crRNA-tracrRNA chimera) was used for identification of possible "nexus" and "hairpins" and the design of sgRNAs for each nuclease with various versions. By shortening the duplex at the upper stem at different locations, the crRNA and tracrRNA were connected with tetra-loop 'gaaa', thus generating possible sgRNA scaffolds (all sgRNA designs of all OMNIs are listed in Table 4). At least two versions of



possible designed scaffolds for each OMNI were synthesized and 5' connected to a 22 nucleotide universal unique spacer sequence (T1, SEQ ID NO: 31149) and were cloned into a bacterial Guide expressing plasmid under a constitutive promoter and a mammalian expressing plasmid under a U6 promoter (pbGuide and pmGuide, accordingly, Table 3).

[0370] In order to overcome potential transcriptional and structural constraints and to assess the plasticity of the sgRNA scaffold in the human cellular environmental context, several versions of sgRNA were tested. In each case, the modifications represent small variations in the nucleotide sequence within the duplex and/or hairpins that were introduced to several synthetic sgRNAs (Table 4).

TABLE-US-00006 T1 (SEQ ID NO: 31149) GGTGCGGTTCCACCAGGGTGTCTG T2 (SEQ ID NO: 31150) ggAAGAGCagagccttGGTCTC

TABLE-US-00007 TABLE 4 Guides OMNI-39 OMNI-40 OMNI-50 OMNI-53 crRNA: trac lower + GUUUUAGUA GUUUUGUUA GUUUGAGAG GUUUGAGAA rRNA duplex upper stem CCUAGAG CCAUAUG UUAUG CCAUG VI of crRNA (SEQ ID (SEQ ID (SEQ ID (SEQ ID NO: 31151) NO: 31161) NO: 31171) NO: 31182) upper + CUUUAGACC UAUAUGACC CAUGACGAG CAUGGUGAG lower stem UACUAAAAU UAACAAAAC UUCAAAU UGCAAAU of tracrRNA (SEQ ID (SEQ ID (SEQ ID (SEQ ID NO: 31152) NO: 31162) NO: 31172) NO: 31183) crRNA: trac lower + GUUUUAGUA GUUUUGUUA GUUUGAGAG GUUUGAGAA rRNA duplex upper stem CCUAGAGAA CCAUAUGAU UUAUGUAA CCAUGUAA V2 of crRNA (SEQ ID U (SEQ ID (SEQ ID (SEQ ID NO: 31153) NO: 31163) NO: 31173) NO: 31184) upper + UUUCUUUAG AUUUUAUUG UUACAUGAC UUACAUGGU lower stem ACCUACUAA ACCUAACAA GAGUUCAAA GAGUGCAA of tracrRNA AAU AAC U (SEQ ID U (SEQ ID (SEQ ID (SEQ ID NO: 31174) NO: 31185) NO: 31154) NO: 31164) tracrRNA AAGGCUUUA AAGGGUUUA AAAAAUUA AAGGAUUUAU (Nexus + UGCCGAGAU UCCCGGACU UUCAACCG CCGAAAUUG hairpins) UAAAGGAUG CGGCUCUUC CCUAUUUAU UAUGCCCGC CCGACGGGC GGAGCCUUU AGGCCGCAG AUUGUGCGG AUCuuuuu UU AUGUUCUGC CAAUAAAAA u (SEQ ID (SEQ ID AUUAUGCUU GGCUCGAAA NO: 31155) NO: 31165) GCUAUUGCA GAGUCUUUU AGCUUUUUU U (SEQ ID (SEQ ID NO: 31186) NO: 31175) Nexus GGCUUUAUG GGGUUUAUC GCCUAUUUA GGAUUAUCC CC CC UAGGC (SEQ ID (SEQ ID (SEQ ID NO: 31156) NO: 31166) NO: 31176) hairpin AAGGAUGCC GGCUCUUCG GCAGAUGUU UGUAUGCCC GACGGGCAU GAGCC CUGCAUUAU GCAUUGUGC CCUUU (SEQ ID GCUUGCUAU GGCAAUAAA (SEQ ID NO: 31167) UGCAAGC AAGGCUCGA NO: 31157) (SEQ ID AAGAGUCUU NO: 31177) U (SEQ ID NO: 31187) sgRNA VI GUUUUAGUA GUUUUGUUA GUUUGAGAG GUUUGAGAA CCUAGAGga CCAUAUGga UUAUGgaaa CCAUGgaaa aaCUUUAGA aaUAUAUGA CAUGACGAG CAUGGUGAG CCUACUAAA CCUAACAAA UUCAAAUAA UGCAAAUAA AUAAGGCUU ACAAGGGUU AAAUUUAUU GGAUUAUCC UAUGCCGAG UAUCCCGGA CAAACCGCC GAAAUUGUA AUUAAAGGA CUCGGCUCU UAUUUAUAG UGCCCGCAU UGCCGACGG UCGGAGCCU GCCGCAGAU UGUGCGGCA GCAUCCUUU UUUU GUUCUGCAU AUAAAAAGG UUU (SEQ ID UAUGCUUGC CUCGAAAGA (SEQ ID NO: 31168) UAUUGCAAG GUCUUUUU NO: 31158) CUUUUUU (SEQ ID (SEQ ID NO: 31188) NO: 31178) sgRNA V2 GUUUUAGUA GUUUUGUUA GUUUGAGAG GUUUGAGAA CCUAGAGAA CCAUAUGAU UUAUGUAAg CCAUGUAAg AgaaaUUUC UgaaaAUUU aaaUUACAU aaaUUACAU UUUAGACCU AUAUGACCU GACGAGUUC GGUGAGUGC ACUAAAAUA AACAAAACA AAUAAAAA AAUAAAGGA AGGCUUUUAU AGGGUUUAU UUUAUUCAA UUAUCCGAA GCCGAGAUU CCCGGACUC ACCGCCUAU AUUGUAUGC AAAGGAUGC GGCUCUUCG UUAUAGGCC CCGCAUUGU CGACGGGCA GAGCCUUUU GCAGAUGUU GCGGCAAUA UCCUUUUUU U (SEQ ID CUGCAUUAU

AAAAGCUC (SEQ ID NO: 31169) GCUUGCUAU GAAAGAGC NO: 31159)  
 UGCAAGCUU UUUUU UUUU (SEQ ID NO: 31189) NO: 31179) Other  
 sgRNA V3 GUUUAAGUA GUUUAGUUA GUUUGAGAG variations CCUAGAGAA  
 CCAUAUGAU UUAUGUgaa AgaaaUUUC UgaaaAUUU aACAUGACG UUUAGACCU  
 AUAUGACCU AGUUCAAA AUUUAAAUA AACUAAACA AAAAAUUUA AGGCUUUUAU  
 AGGGUUUAU UUCAACCG GCCGAGAUU CCCGGACUC CCUAUUUAU AAAGGAUGC  
 GGCUCUUCG AGGCCGCAG CGACGGGCA GAGCCUUUU AUGUUCUGC UCCUUUUUU  
 U (SEQ ID AUUAUGCUU (SEQ ID NO: 31170) GCUAUUGCA NO: 31160)  
 AGCUUUUUU (SEQ ID NO: 31180) sgRNA V4 GUUUGAGAG UUAUGUAga  
 aaUACAUGA CGAGUUCAA AUAAAAAUU UAUUCAAC CGCCUAUUU AUAGGCCGC  
 AGAUGUUCU GCAUUAUGC UUGCUAUUG CAAGCUUUU UU (SEQ ID NO: 31181)  
 TABLE-US-00008 TABLE 5 PAMs OMNI- OMNI- OMNI- OMNI- 39 40 50 53 Bacterial PAM  
 NNGYAD NYGRV NRTA Depletion General PAM NNGYAA NYGAV NRTA Specific Activity  
 (1- 0.99 0.95 1.00 Depletion score)\* TXTL PAM NNGHAD NYGRV NGG NRHR Depletion  
 General PAM NNGYAA NYGRV NGG NAWA Specific Activity (1- 0.95 0.97 0.98 0.97 Depletion  
 score)\* sgRNA V2 V2 V2 V2 mammalian VTGAAG NGG NRTA system \*Depletion score -  
 Average of the ratios from two most depleted sites \*\* Depends on OMNI expression and sgRNA  
 transcription

#### Bacterial PAM Depletion Assay.

[0371] To confirm that each of the identified loci are functional CRISPR-OMNI nuclease systems and to identify their PAMs, *E. coli* strain BW25141 (IDE3) was co-transformed with: (1) a library of plasmid pool containing a randomized PAM sequences of 8 N's flanking a unique protospacer (pbPOS 12 library, Table 3), (2) plasmids encoding *E. coli* codon-optimized OMNI nucleases, pbNNC2 (Table 3) and (3) a plasmid encoding designed sgRNA targeting the protospacer of the library or a non-targeting gRNA as control (pbGuide, T2 and T1 respectively, Table 3). Next, cells were selected for all three plasmids by recovering them on media containing appropriate antibiotics. In this assay, plasmids containing a PAM are cleaved and the cells that contain them cannot grow, while cells containing plasmids with non-PAMs are able to propagate. A survived plasmid DNA pool was isolated, and the library was sequenced using a 75-cycle NextSeq kit (Illumina). PAM representation in the library was determined using a custom script and compared between OMNI and control. By comparing the frequency of a sequence in the library after selection of the targeting guide (T2) relative to the non-targeting (T1), individual PAM sequences could be identified. The presented data reflect a condensed 4N window library with all possible locations along the 8 bp sequence. Sequence motifs were generated using the Weblogo tool (weblogo.berkeley.edu). Activity of the OMNI nuclease was estimated based on the average of the two most depleted sequences and was calculated as: 1-Depletion score (Depletion score-Average of the ratios from two most depleted sites). OMNI nucleases with scores that are higher than 0.6 were considered as active. Following deep sequencing we detected depletion in the tested OMNI systems, indicating functional DNA interference in a heterologous host (Table 5).

#### In-Vitro Depletion Assay by TXTL.

[0372] Depletion of PAM sequences in-vitro was followed according to Maxwell et al, Methods. 2018. Briefly, linear DNA expressing the OMNI nucleases and the sgRNAs under a T7 promoter were added to a TXTL mix (Arbor Bioscience) together with a linear construct expressing T7 polymerase. RNA expression and protein translation by the TXTL mix results in the formation of the RNP complex. Since linear DNA was used, Chi6 sequences, a RecBCD inhibitor, were added to protect the DNA from degradation. The sgRNA spacer is designed to target a library of plasmids containing the targeting protospacer (pbPOS 12 library, Table 3) flanked by an 8N randomized set of potential PAM sequences. Depletion of PAM sequences from the library is measured by adding the adapters and indices necessary for high-throughput sequencing using PCR to both the cleaved library and to a control library expressing a non-targeting gRNA (T1). Following deep sequencing,

the activity was confirmed in-vitro by depletion in 10 out of 10 tested OMNI, showing similar PAM pattern as discovered in *E. coli* (Table 5). Thus, the in-vitro activity displayed by OMNI nucleases indicated functional DNA interference by in-vitro system (Table 5).

#### Pam Library in Mammalian System

[0373] While a PAM sequence preference is considered an inherent property of the nuclease, it might be effected, to some extent, from the cellular environment and genomic composition and size. Since the human cellular environment is significantly different from the bacterial one with respect to those properties, a “fine tuning” step was introduced to address potential differences in PAM preferences in human cellular context. To this end, a PAM library was constructed in human cell line. In this assay, The PAM library is introduced to the cells using a viral vector, as constant target sequence followed by a stretch of 6× N's. Upon introduction of OMNI and sgRNA targeting the library constant target site, NGS analysis is used to identify the edited sequences and the PAM associated with them. The enriched edited sequences are then used to define the PAM consensus. This method is applied to determine the “human” PAM requirements of OMNI nuclease (Table 5, “mammalian system”). OMNI-50 PAM is identical using TXTL or mammalian system. OMNI-53 PAM is a reduced version of the PAM identified by TXTL. On the other hand OMNI-40 show a more strict PAM compared with TXTL results. OMNI-39 PAM could not be determined using the mammalian system due to low number of editing events.

#### Expression of OMNI Nucleases Coded by an Optimized DNA Sequence in Mammalian Cells

[0374] First, expression of each of the optimized DNA sequences coding for each of the OMNI proteins OMNI-39, OMNI-40, OMNI-50, and OMNI-53 in mammalian cells, was validated. To this end, an expression vector coding for an HA-tagged OMNI nuclease or *Streptococcus pyogenes* Cas9 (SpCas9) linked by P2A peptide to mCherry (pmOMNI, Table 3) was introduced into HeLa cells using the Jet-optimus™ transfection reagent (polyplus-transfection). P2A peptide is a self-cleaving peptide which can induce the cleaving of the recombinant protein in cell, such that the OMNI nucleases and the mCherry are separated upon expression and the mCherry can serve as indicator for transcription efficiency of the OMNI from the expression vector. The level of expression for each OMNI in the activity assays was determined using flow cytometry.

#### Activity in Human Cells on Endogenous Genomic Targets

[0375] OMNI were also assayed for their ability to promote editing on specific genomic locations in human cells. To this end, per each OMNI the corresponding OMNI-P2A-mCherry expression vector (pmOMNI, Table 3) was transfected into HeLa cells together with sgRNA designed to target a specific location in the human genome (pmGuide, Table 3). At 72h, cells were harvested, and half were used for quantification of transfection efficiency by FACS using mCherry fluorescence as marker. The rest of the cells were lysed, and their genomic DNA content was used in PCR reaction, amplifying the corresponding putative genomic targets. Amplicons were subjected to NGS and the resulting sequences were then used calculate the percentage of editing events in each target site. Short insertions or deletions (Indels) around the cut site are the typical outcome of repair of DNA ends following nuclease induced DNA cleavage. The calculation of % editing was therefore deduced from the fraction of Indels containing sequences within each amplicon. All editing values were normalized to the transfection and translation efficacy obtained for each experiment and deduced from the percentage of mCherry expressing cells. The normalized values represent the effective editing levels within the population of cells that expressed the nucleases.

[0376] Genomic activity of each OMNI was assessed using a panel of 4 to 11 unique sgRNA each design to target a different genomic location. The results of these experiments are summarized in Table 6. As can be seen in the table (column 6, “% editing”), all of the OMNI exhibit high and significant editing levels compared to the negative control (column 9, “% editing in neg control”) in all or most target site tested. OMNI-39 exhibited high and significant editing levels in 2/4 sites tested, OMNI-40 exhibit high and significant editing Levels in ¾ sites tested, OMNI50 exhibited high and significant editing levels in 11/11 sites tested, and OMNI-53 exhibited high and

significant editing levels in 3/4 sites tested.

TABLE-US-00009 TABLE 6 Activity in human cells on endogenous genomic targets 3'(PAM containing) genomic Corresponding seq Genomic Spacer Spacer (PAM = % trans- Nuclease site name sequence Bold) % indels fection OMNI-39 CXCR4 CXCR4g1\_OMNI39 CCAAGUG **TGGCAAGA** 49.8-73.2 67.13 site 1 AUAACA CGAGGA (SEQ ID NO: 31190) EMX1 EMX1g1\_OMNI39 GUCACCU **GGGCAACC** 3.9-6.3 site 1 CCAUGA CUAGGGU (SEQ ID NO: 31191) EMX1 EMX1g2\_OMNI39 GCCGCCA **AAGCAATG** 22.8-54.7 site 2 UUGACAG AGGGAC (SEQ ID NO: 31192) PDCD1 PDCD1g1\_OMNI39 AACUGGU **CAGCAACC** 3.71 73.67 site 1 ACCGCAU GAGCCC (SEQ ID NO: 31193) OMNI-40 EMX1 EMX1g1\_OMNI40 CAUCAGG **CTGAGTGT** 25-37.5 50.33 site 1 CUCUCAG CUCAGC (SEQ ID NO: 31194) CXCR4 CXCR4g2\_OMNI40 AGGUGCC **CTGACACT** 0.20 53.60 site 2 GUUUGUU CAUUUU (SEQ ID NO: 31195) PDCD1 PDCD1g1\_OMNI40 CCAGUUG **ACGACTGG** 23.59 28.33 site 2 UAGCACC GCCCAG (SEQ ID NO: 31196) PDCD1 PDCD1g2\_OMNI40 UCUCCCC **GTGACCGA** 16.66 49.00 site 3 AGCCCUG CUCGUG (SEQ ID NO: 31197) OMNI-50 EMX1 EMX1g1\_OMNI50 UCUGUGA **GGGAGCAG** 44.18-25.72 site 2 AUGUUAG ACCCAU (SEQ ID NO: 31198) EMX1 EMX1g2\_OMNI50 CCAUGGG **AGGGGACC** 55.81 site 3 AGCAGCU GGUCAG (SEQ ID NO: 31199) CXCR4 CXCR4g1\_OMNI50 GCAAGAG **AGGAGCGC** 29.58-32.14 site 3 ACCCACA CACCGG (SEQ ID NO: 31200) CXCR4 CXCR4g2\_OMNI50 ACACCGG **TGGGGGAG** 42.13-49.85 site 4 AGGAGCG CCCGCU (SEQ ID NO: 31201) PDCD1 PDCD1g1\_OMNI50 CGUCUGG **TGGGCTGG** 13.35-8.7 site 4 GCGGUGC UACAAC (SEQ ID NO: 31202) PDCD1 PDCD1g2\_OMNI50 CUACAAC **AGGATGGT** 17.53 site 5 UGGGCUG GCGGCC (SEQ ID NO: 31203) ELANE ELANEg35\_OMNI50 AGUCCGG **GGGGAGCA** 40.92-55.39 g35 GCUGGGA GCGGGU (SEQ ID NO: 31204) ELANE ELANEg58\_OMNI50 GCUGCGG **TGGGACTC** 11.11 20.50 g58 GAAAGGG AUUCCC (SEQ ID NO: 31205) ELANE ELANEg38\_OMNI50 ACAGCGG **GGGGGACG** 9.99 g38 GUGUAGA CUCCGA (SEQ ID NO: 31206) ELANE ELANEg39\_OMNI50 CAGCGGG **GGGGACGT** 24.87 g39 UGUAGAC UCCGAG (SEQ ID NO: 31207) ELANE ELANEg62\_OMNI50 GUCAAGC **GGGACAGA** 38.87-52.74 g62 CCCAGAG GCCACA (SEQ ID NO: 31208) OMNI-53 EMX1 EMXg1\_OMNI53 GCCUGGG **TGTAGCCT** 18.3-36.7 49.63 site 4 GCCCCUA ACCCUA (SEQ ID NO: 31209) CXCR4 CXCR4g2\_OMNI53 AUUUUCU **AATATACC** 14.1-12.5 38.33 site 5 GACACUC CCGCCC (SEQ ID NO: 31210) PDCD1 PDCD1g1\_OMNI53 AUCCUGG **TGTAGCAC** 11.5 51.27 site 6 CCGCCAG CCCAGU (SEQ ID NO: 31211) PDCD1 PDCD1g2\_OMNI53 GGAGAGC **GGTACCGC** 1.93 30.30 site 7 UUCGUGC UAAACU (SEQ ID NO: 31212) % editing % trans- Norm. % Genomic Norm. % in neg fection in editing in Nuclease site editing control neg control neg control OMNI-39 CXCR4 0.08 76.70 0.107791557 site 1 EMX1 site 1 EMX1 site 2 PDCD1 5.03 0.07 76.70 0.086641622 site 1 OMNI-40 EMX1 0.12 53.37 0.231979262 site 1 CXCR4 0.37 0.21 53.37 0.396940137 site 2 PDCD1 83.25 0.09 53.37 0.174121445 site 2 PDCD1 34.00 0.01 0.18 8.107932801 site 3 OMNI-50 EMX1 0.02 site 2 EMX1 site 3 CXCR4 0.18 site 3 CXCR4 0.22 site 4 PDCD1 0.05 site 4 PDCD1 site 5 ELANE 0.24 5.95 3.982225429 g35 ELANE 54.23 0.18 5.95 2.974553445 g58 ELANE g38 ELANE g39 ELANE 0.12 5.95 2.002503126 g62 OMNI-53 EMX1 0.15 43.80 0.333213614 site 4 CXCR4 0.22 43.80 0.509942217 site 5 PDCD1 22.50 0.05 43.80 0.105935337 site 6 PDCD1 6.38 0.01 43.80 0.019429028 site 7

### Intrinsic Fidelity in Human Cells

[0377] The intrinsic fidelity of a nuclease is a measure of its cleavage specificity. High fidelity nuclease is a nuclease that promotes cleavage on the intended target (“on-target”) with minimal or no cleavage of the unintended target (“Off-target”). In CRISPR nucleases, the target is acquired based on sequence complementarity to the spacer element of the guide RNA. Off-targeting results

from similarity of an unintended target to the spacer sequence. The intrinsic fidelity of OMNIs at the genomic level in human cells was measured by conducting an activity assay, followed by PCR amplification, NGS, and InDel analysis for both the on-target region and a pre-validated off target region. A measurement of intrinsic fidelity for OMNI-50 is provided in FIG. 2. In this example, OMNI-50 fidelity was measured using two guide RNAs independently, in each case a side by side measurement of SpCas9 is provided for reference. The first site was targeted using the ELANE g35 gRNA (Table 6) which has a defined on-target site upstream to the ELANE gene on chr19 and an off-target site on chr15. As can be seen in FIG. 2, the on/off target editing efficiency ratio obtained by OMNI-50 was 41:0 while SpCas9 on/off ratio is 6.8:1 (40.9%/0%; 18.6%/2.7% respectively). The second site was targeted by ELANE g62 gRNA (Table 6). This gRNA spacer sequence has a defined on-target at the ELANE gene on chr19 and an off-target site on chr1. In this case, the On/Off ratio obtained by OMNI-50 was 72:1 compared to 1.7:1 ratio obtained by SpCas9 (38.9%/0.6%; 43.1%/625.8% respectively). These results demonstrate that OMNI-50 has a significantly higher intrinsic fidelity in comparison to SpCas9 in human cellular environment.

**Purification of OMNIs Protein.**

[0378] The OMNI-50 open-reading frame was cloned into bacterial expression plasmids (T7-NLS-OMNI-NLS-HA-His-tag, pET9a, Table 3) and expressed in C43 cells (Lucigen). Cells were grown in Terrific Broth to mid-log phase and the temperature was lowered to 18° C. Expression was induced at 0.6 OD with 1 mM IPTG for 16-20 h before harvesting and freezing cells at -80° C. Cell paste was resuspended in lysis buffer (50 mM NaH.sub.2PO.sub.4, 300 mM NaCl, 10 mM imidazole pH8.0, 1 mM TCEP) supplemented with EDTA-free complete protease inhibitor cocktail set III (Calbiochem). Cells were lysed using sonication and cleared lysate was incubated with Ni-NTA resin. The resin was loaded onto gravity column and washed with wash buffer (50 mM NaH.sub.2PO.sub.4, 300 mM NaCl, 50 mM imidazole pH8.0, 1 mM TCEP) and OMNI protein eluted with wash buffer supplemented with 100-500 mM Imidazole. Fractions containing OMNI protein were pooled and concentrated and loaded onto a centricon (Amicon Ultra 15 ml 100K, Merck), and buffer exchanged to GF buffer (50 mM Tris-HCl pH 7.5, 500 mM NaCl, 10% glycerol, 0.4M Arginine). The concentrated OMNI protein was further purified by SEC on HiLoad 16/600 Superdex 200 pg-SEC, AKTA Pure (GE Healthcare Life Sciences) with a 50 mM Tris-HCl pH 7.5, 500 mM NaCl, 10% glycerol, 0.4M Arginine. Fractions containing OMNI protein were pooled and concentrated and loaded onto a centricon (Amicon Ultra 15 ml 100K, Merck) with a final storage buffer of 10 mM Tris-HCl, pH 7.5, 150 mM NaCl, 10% glycerol and 1 mM TCEP. Purified OMNI protein was concentrated to 10 mg/ml stocks and flash-frozen in liquid nitrogen and stored at -80° C.

**Guide Optimization by RNP Activity Assay.**

[0379] Synthetic sgRNAs of OMNI-50 were synthesized with three 2'-O-methyl 3'-phosphorothioate at the 3' and 5' ends (Agilent). Activity assay of OMNI-50 RNP with different spacer lengths (17-23 nucleotides) of guide 35 (Table 7, FIG. 3A). Briefly, 4 pmol of OMNI-50 nuclease were mixed with 6 pmol of synthetic guide. After 10 min of incubation at room-temp. the RNP complexes were reacted with 100 ng template of On-target. Only spacer larger than 22 nucleotides show near full cleavage of the On-target template. Furthermore, reducing amounts of RNPs (4, 2, 1.2, 0.6 and 0.2 µmol) with spacer lengths 20-23 were reacted with 100 ng DNA target template (FIG. 31D) Spacer at lengths of 22 nucleotides showed better cleavage activity even at lower RNP concentrations.

TABLE-US-00010

| TABLE               | 7 synthetic sgRNA   | (RNP) Name | O50_ELANE_V2_g35_23                     |
|---------------------|---------------------|------------|-----------------------------------------|
| O50_ELANE_V2_g35_22 | O50_ELANE_V2_g35_21 | Spacer     | UgcAGUCCG gcAGUCCGG                     |
| cAGUCCGGG           | GGCUGGGAG           | GCUGGGAGC  | CUGGGAGCG CGGGU GGGU (SEQ               |
| GGU(SEQ             | (SEQ                | ID ID      | NO: ID NO: NO: 31213) 31217) 31219)     |
| Scaffold            | gUUUGAGAG           | gUUUGAGAG  | gUUUGAGAG UUAUGUAAg UUAUGUAAg UUAUGUAAg |
| aaaUUACAU           | aaaUUACAU           | GACGAGUUC  | GACGAGUUC GACGAGUUC AAAUAAAAA           |

AAUAAA AAAUAAA AAAUAAA AAAUAAA AAAUAAA ACCGCCUAU  
ACCGCCUAU ACCGCCUAU UUAUAGGCC UUAUAGGCC UUAUAGGCC GCAGAUGUU  
GCAGAUGUU GCAGAUGUU CUGCAUUAU CUGCAUUAU CUGCAUUAU GCUUGCUAU  
GCUUGCUAU GCUUGCUAU UGCAAGCUU UGCAAGCUU UGCAAGCUU UUUU (SEQ  
UUUU (SEQ ID NO: ID NO: ID NO: 31179) 31179) 31179) Version V2 V2  
V2 Full sgRNA UgCAGUCCG gcAGUCCGG cAGUCCGGG sequence GGCUGGGAG  
GCUGGGAGC CUGGGAGCG CGGGUgUUU GGGUgUUUG GGUgUUUGA GAGAGUUAU  
AGAGUUAUG GAGUUAUGU GUAAGaaaU UAAGaaaUU AAgaaaUUA UACAUGACG  
ACAUGACGA CAUGACGAG AGUUCAAA AUUCAAAUA UUCAAAUAAA AAAAAUUUA  
AAAAUUUAU AAAUUUAU UUCAAAACCG UCAAACCGC CAAACCGCC CCUAUUUAU  
CUAUUUUAU UAUUUUAU AGGCGCGCAG GGCCGCAGA GCCGCAGAU AUGUUCUGC  
UGUUCUGCA GUUCUGCAU AUUAUGCUU UUAUGCUUG UAUGCUUGC GCUAUUGCA  
CUAUUGCAA UAUUGCAAG AGCUUUUUU GCUUUUUU CUUUUUU (SEQ ID (SEQ  
ID (SEQ ID NO: 31214) NO: 31218) NO: 31220) Protospacer CTGTTGCTG  
CTGTTGCTG CTGTTGCTG (with PAM = CAGTCCGGG CAGTCCGGG CAGTCCGGG  
Bold) - On CT**GGG**AGCG CT**GGG**AGCG CT**GGG**AGCG target GGTGGGGAG  
GGTGGGGAG GGTGGGGAG CAGAGGG CAGAGGG CAGAGGG (SEQ ID (SEQ ID  
(SEQ ID NO: 31215) NO: 31215) NO: 31215) Protospacer GTTAAGAg GTTAAGAg  
GTTAAGAg (with PAM = CAGTCCaGG CAGTCCaGG CAGTCCaGG Bold) - Off  
CTGGGAGCa CTGGGAGCa CTGGGAGCa target GGT**GGG**GAG GGT**GGG**GAG  
GGT**GGG**GAG AGGAGGG AGGAGGG AGGAGGG (SEQ ID (SEQ ID (SEQ ID NO:  
31216) NO: 31216) NO: 31216) Name O50\_ELANE\_V2\_g35\_20 O50\_ELANE\_V2\_g35\_19  
Spacer AGUCCGGGC GUCCGGGCU UGGGAGCGG GGGAGCGGG GU (SEQ U (SEQ  
ID ID NO: NO: 31223) 31221) Scaffold gUUUGAGAG gUUUGAGAG UUAUGUAAg  
UUAUGUAAg aaaUUACAU aaaUUACAU GACGAGUUC GACGAGUUC AAAUAAAAA  
AAAUAAAAA UUUAUUCAA UUUAUUCAA ACCGCCUAU ACCGCCUAU UUAUAGGCC  
UUAUAGGCC GCAGAUGUU GCAGAUGUU CUGCAUUAU CUGCAUUAU GCUUGCUAU  
GCUUGCUAU UGCAAGCUU UGCAAGCUU UUUU (SEQ UUUU (SEQ ID NO: ID  
NO: 31179) 31179) Version V2 V2 Full sgRNA AGUCCGGGC GUCCGGGCU sequence  
UGGGAGCGG GGGAGCGGG GUgUUUGAG UgUUUGAGA AGUUAUGUA GUUAUGUAA  
AgaaaUUAC gaaaUUACA AUGACGAGU UGACGAGUU UCAAAUAAA CAAAUAAAA  
AAUUUAUUC AUUUUAUUC AAACCGCCU AACCGCCUA AUUUUAUAGG UUUAUAGGC  
CCGCAGAU CCGCAGAU UUCUGCAU UCUGCAUUA AUGCUUGCU UGCUUGCUA  
AUUGCAAGC UUGCAAGCU UUUUUU UUUUU (SEQ ID (SEQ ID NO: 31222) NO:  
31224) Protospacer CTGTTGCTG CTGTTGCTG (with PAM = CAGTCCGGG  
CAGTCCGGG Bold) - On CT**GGG**AGCG CT**GGG**AGCG target GGTGGGGAG  
GGTGGGGAG CAGAGGG CAGAGGG (SEQ ID (SEQ ID NO: 31215) NO: 31215)  
Protospacer GTTAAGAg GTTAAGAg (with PAM = CAGTCCaGG CAGTCCaGG  
Bold) - Off CTGGGAGCa CTGGGAGCa target GGT**GGG**GAG GGT**GGG**GAG  
AGGAGGG AGGAGGG (SEQ ID (SEQ ID NO: 31216) NO: 31216) Name  
O50\_ELANE\_V2\_g35\_18 O50\_ELANE\_V2\_g35\_17 Spacer UCCGGGCUGGG  
CCGGGCUGGGA AGCGGGU GCGGGU (SEQ (SEQ ID NO: ID NO: 31225) 31227)  
Scaffold gUUUGAGAGUU gUUUGAGAGUU AUGUAAgaaaU AUGUAAgaaaU  
UACAUGACGAG UACAUGACGAG UUCAAAUAAAA UUCAAAUAAAA  
AUUUUAUUC AUUUUAUUC CCGCCUAUUUA CCGCCUAUUUA UAGGCCGCAGA  
UAGGCCGCAGA UGUUCUGCAU UGUUCUGCAU AUGCUUGCUAU  
AUGCUUGCUAU UGCAAGCUUUU UGCAAGCUUUU UU (SEQ ID UU (SEQ ID  
NO: 31179) NO: 31179) Version V2 V2 Full sgRNA UCCGGGCUGGG CCGGGCUGGGA  
sequence AGCGGGUgUUU GCGGGUgUUUG GAGAGUUAUGU AGAGUUAUGUA  
AAGaaaUUACA AgaaaUUACAU UGACGAGUUA GACGAGUUA AUAUAAAAUUU



levels, and 21-23 nt spacers resulted in the most significant editing levels, in agreement with the in vitro and the U2OS experiment described above.

#### Example: ELANE Mutant Targeting Analysis

[0383] Guide sequences comprising 17-30 contiguous nucleotides containing nucleotides in the sequence set forth in any one of SEQ ID NOs: 1-31117 are screened for high on target activity using SpCas9 in HeLa cells. On target activity is determined by DNA capillary electrophoresis analysis.

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## Claims

1. An ex vivo or in vitro method for modifying in a cell a mutant allele of the elastase, neutrophil expressed (ELANE) gene wherein the mutant allele has a mutation associated with severe congenital neutropenia (SCN) or cyclic neutropenia (CyN), the method comprising introducing to

the cell a composition comprising: (a) a CRISPR nuclease, or a DNA polynucleotide encoding said CRISPR nuclease, wherein said CRISPR nuclease comprises: (i) the amino acid sequence of SEQ ID NO: 31125; or (ii) the amino acid sequence encoded by the nucleotide sequence of SEQ ID NO: 31131 or 31132; and (b) a first RNA molecule, or a DNA polynucleotide encoding the first RNA molecule, wherein said first RNA molecule comprises: (i) a CRISPR RNA (crRNA) molecule and a transactivating CRISPR RNA (tracrRNA) molecule; or (ii) a single-guide RNA (sgRNA molecule), wherein the first RNA molecule comprises (I) a guide sequence portion comprising 20, 21, or 22 contiguous nucleotides of the sequence set forth in SEQ ID NO: 31213 or (II) a guide sequence portion comprising SEQ ID NO: 31221, 31219, 31217, or 31213, wherein the CRISPR nuclease and the first RNA molecule form a complex, and wherein said complex binds to the mutant allele of the ELANE gene and introduces a double strand break in the mutant allele of the ELANE gene.

**2.** The method of claim 1, further comprising introduction of a second RNA molecule, or a DNA polynucleotide encoding the second RNA molecule, wherein the second RNA molecule comprises (i) a crRNA molecule and a tracrRNA molecule or (ii) an sgRNA molecule, wherein the second RNA molecule and the CRISPR nuclease form a complex, and wherein the complex of the second RNA molecule and CRISPR nuclease introduces a second double strand break in the mutant allele of the ELANE gene.

**3.** The method of claim 2, wherein the second double strand break is within a non-coding region of the ELANE gene.

**4.** The method of claim 1, comprising obtaining a cell with a mutant allele of the ELANE gene associated with severe congenital neutropenia (SCN) or CyN from a subject with a mutant allele of the ELANE gene associated with SCN or CyN and/or suffering from SCN or CyN.

**5.** The method of claim 4, comprising obtaining the cell from the subject by (i) mobilization and apheresis, (ii) apheresis, (iii) mobilization, or (iv) bone marrow aspiration.

**6.** The method of claim 4, wherein cytokine is introduced to the cell prior to introducing the composition to the cell.

**7.** The method of claim 4, wherein the cell is expanded by culturing the cell to obtain a larger quantity of the cells, and wherein the cells are cultured with one or more of: stem cell factor (SCF), interleukin 3 (IL-3), and granulocyte-macrophage colony-stimulating factor (GM-CSF); wherein the cells are cultured with at least one cytokine.

**8.** The method of claim 7, wherein the at least one cytokine is a recombinant human cytokine.

**9.** The method of claim 1, wherein the first RNA molecule comprises: (a) a crRNA molecule and a tracrRNA molecule, wherein the tracrRNA molecule comprises the sequence set forth in SEQ ID NO: 31178, 31179, 31180, or 31181; or (b) an sgRNA molecule, wherein the sgRNA molecule comprises the sequence set forth in SEQ ID NO: 31222, 31229, or 31230.

**10.** The method of claim 9, further comprising introduction of a second RNA molecule, or a DNA polynucleotide encoding the second RNA molecule, wherein the second RNA molecule comprises (i) a crRNA molecule and a tracrRNA molecule or (ii) an sgRNA molecule, wherein the second RNA molecule and the CRISPR nuclease form a complex, and wherein the complex of the second RNA molecule and CRISPR nuclease introduces a second double strand break in the mutant allele of the ELANE gene.

**11.** The method of claim 10, wherein the second double strand break is within a non-coding region of the ELANE gene.

**12.** The method of claim 9, comprising obtaining a cell with a mutant allele of the ELANE gene associated with severe congenital neutropenia (SCN) or CyN from a subject with a mutant allele of the ELANE gene associated with SCN or CyN and/or suffering from SCN or CyN.

**13.** The method of claim 12, wherein the cell is expanded by culturing the cell to obtain a larger quantity of the cells, and wherein the cells are cultured with one or more of: stem cell factor (SCF), interleukin 3 (IL-3), and granulocyte-macrophage colony-stimulating factor (GM-CSF); wherein the cells are cultured with at least one cytokine.

- 14.** The method of claim 13, wherein the at least one cytokine is a recombinant human cytokine.
- 15.** An ex vivo or in vitro method for modifying in a cell a mutant allele of the elastase, neutrophil expressed (ELANE) gene wherein the mutant allele has a mutation associated with severe congenital neutropenia (SCN) or cyclic neutropenia (CyN), the method comprising introducing to the cell a composition comprising: (c) a CRISPR nuclease, or a DNA polynucleotide encoding said CRISPR nuclease, wherein said CRISPR nuclease comprises: (iii) the amino acid sequence of SEQ ID NO: 31125; or (iv) the amino acid sequence encoded by the nucleotide sequence of SEQ ID NO: 31131 or 31132; and (d) a first RNA molecule, or a DNA polynucleotide encoding the first RNA molecule, wherein said first RNA molecule comprises: (iii) a CRISPR RNA (crRNA) molecule and a transactivating CRISPR RNA (tracrRNA) molecule; or (iv) a single-guide RNA (sgRNA molecule), wherein the first RNA molecule comprises a guide sequence portion comprising the sequence set forth in SEQ ID NO: 31205, 31206, 31207, or 31208, wherein the CRISPR nuclease and the first RNA molecule form a complex, and wherein said complex binds to the mutant allele of the ELANE gene and introduces a double strand break in the mutant allele of the ELANE gene.
- 16.** The method of claim 15, further comprising introduction of a second RNA molecule, or a DNA polynucleotide encoding the second RNA molecule, wherein the second RNA molecule comprises (i) a crRNA molecule and a tracrRNA molecule or (ii) an sgRNA molecule, wherein the second RNA molecule and the CRISPR nuclease form a complex, and wherein the complex of the second RNA molecule and CRISPR nuclease introduces a second double strand break in the mutant allele of the ELANE gene.
- 17.** The method of claim 16, wherein the second double strand break is within a non-coding region of the ELANE gene.
- 18.** The method of claim 15, comprising obtaining a cell with a mutant allele of the ELANE gene associated with severe congenital neutropenia (SCN) or CyN from a subject with a mutant allele of the ELANE gene associated with SCN or CyN and/or suffering from SCN or CyN.
- 19.** The method of claim 18, comprising obtaining the cell from the subject by (i) mobilization and apheresis, (ii) apheresis, (iii) mobilization, or (iv) bone marrow aspiration.
- 20.** The method of claim 18, wherein cytokine is introduced to the cell prior to introducing the composition to the cell.
- 21.** The method of claim 18, wherein the cell is expanded by culturing the cell to obtain a larger quantity of the cells, and wherein the cells are cultured with one or more of: stem cell factor (SCF), interleukin 3 (IL-3), and granulocyte-macrophage colony-stimulating factor (GM-CSF); wherein the cells are cultured with at least one cytokine.
- 22.** The method of claim 21, wherein the at least one cytokine is a recombinant human cytokine.
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